

A Singular Fold Model for Capacity-Constrained Dynamics

James Kovalenko james@progressfunction.org June 2026 *Preprint. Not peer reviewed.*

Abstract

We study a minimal singular fast-slow dynamical model for capacity-constrained compounding. The fast subsystem takes the smooth saddle-node normal form, while the slow vector field $g(x) = \alpha - \beta k/x$ is singular at the boundary $x = 0$.

Our results are local and explicit: a saddle-node bifurcation of the fast subsystem (Proposition 1); a conditional friction-scaling law $F \sim (Q - V)^{-1/2}$ along the attracting branch under the coupling $F(x) = k/x$ (Proposition 2); a barrier monotonicity of $L(V) = -F(x^*(V))$ on the congestion regime of the reduced slow flow (Proposition 3); a boundedness equivalence between bounded friction and bounded-away-from-fold trajectories on the reduced flow (Criterion 1); a fold-repulsion and exit-time theorem for the full singular system on strict congestion subsets at *every* fixed fold distance $\delta > 0$ (Theorem 1), with an explicit smallness threshold $\varepsilon \leq \varepsilon_0(\delta, r)$; and an isolating-block certificate covering all initial conditions in the slow-manifold tube (Proposition 5). Composing the two (Corollary 1): for ε below the stated Lyapunov and isolating-block thresholds ($\varepsilon \leq \varepsilon_0(\delta, r)$ and $\varepsilon < \varepsilon_{\text{blk}}$, both explicit), every congestion-tube trajectory exits the cooperative face without fold approach, in explicit time with exact leading term $16/(\gamma_0 \mu_0 \delta \varepsilon)$. The minimum-distance threshold $\delta_{\min}(r)$ is an artifact of the Lyapunov normalization and is removed by a weighted construction.

We do not prove a global sufficiency result on the full singular system, a Krupa–Szmolyan classification, or uniform-in- δ behavior as $\delta \rightarrow 0$ at fixed ε (the fold neighborhood in the joint limit). The blow-up regularisation of the $1/x$ singularity is not completed here – but all three chart legs are now proven: the inner problem is solved in closed form, the full chart atlas and transition maps are derived, and the entry leg (the K_1 centre manifold), the passage leg (K_2 slaving, Gaussian transient, α -correction), and the exit leg (the landing on P_1 ’s centre manifold, which un-blows to the classical first-order slow manifold) are proved by conventional invariant-manifold arguments, with every computation machine-checked (§6.2), and a subset of the finite certificate algebra now carries Lean 4 kernel proofs (Appendix E). The gluing of the legs and the time bookkeeping remain formal sketches, and we record exactly what is left for a rigorous resolution.

We derive four fold-consistent CSD predictions for exogenous approach and one barrier prediction (P5) that tests the autonomous certificate directly, with explicit identifiability assumptions and competing mechanisms, and specify a falsification protocol.

Contents

1. Status of Claims
 2. Introduction
 3. Local Model: Fast Subsystem
 4. Reduced Slow Flow on the Attracting Branch
 5. Fold Repulsion in the Full System
 6. The Singularity and Its Open Resolution
 7. Empirical Predictions and Identifiability
 8. Discussion
- Appendix A: Empirical Protocol for Testing P4
 - Appendix B: Synthetic Validation
 - Appendix C: Full-System Simulations
 - Appendix D: Explicit Thresholds for Theorem 1
 - Appendix E: Kernel-Verified Certificate Layer (Lean 4)
-

1. Status of Claims

We separate proved local results from open singular-perturbation items.

Claim	Status
Saddle-node bifurcation of the fast subsystem (Prop. 1)	Proved. Standard normal-form verification.
Friction scaling $F \sim (Q - V)^{-1/2}$ on attracting branch (Prop. 2)	Proved , conditional on coupling $F(x) = k/x$.
Barrier monotonicity on the congestion regime of the reduced flow (Prop. 3)	Proved on the reduced slow flow restricted to the attracting branch.
Boundedness equivalence (Crit. 1)	Proved. Equivalence between bounded friction and bounded-away-from-fold on the reduced flow.
Fold-repulsion funnel and exit-time bound on strict congestion subsets of the full singular system (Thm. 1)	Proved for every fixed $\delta > 0$, with $\varepsilon \leq \varepsilon_0(\delta, r)$. Weighted Lyapunov certificate; the $\delta_{\min}(r)$ threshold is removed (Remark 5).
Isolating-block certificate: tube exit only through the cooperative-side face (Prop. 5)	Proved. Face-flux computation; covers all initial conditions in the tube.
Rated exit for all tube initial conditions (explicit exit time)	Proved (Theorem 1 + Proposition 5; Corollary 1).
Uniform-in- δ certification at fixed ε (joint limit $\delta \rightarrow 0$; fold neighborhood)	Open. $\varepsilon_0(\delta, r) \rightarrow 0$ as $\delta \rightarrow 0$; requires blow-up resolution; mechanism identified in the inner chart (§6); all three chart legs now proven (Lemmas 2–4, §6.2), the gluing still open.
Local asymptotic stability of the cooperative equilibrium (x_0, V_0)	Proved (Proposition 4). Hurwitz linearisation plus Lyapunov indirect method.
Explicit overlap between Proposition 4’s basin and Theorem 1’s certified region across $V = V_0 + r$	Open. Quantitative matching requires Lyapunov-function scaling not pursued here.
Krupa–Szmolyan jump-point classification	Open (inputs assembled). Standard hypotheses fail at $x = 0$; the chart-leg lemmas (Lemmas 2–4, §6.2) now supply the centre-manifold inputs at E_1 and P_1 , leaving the gluing and the time bookkeeping.
Full-divisor blow-up regularisation	Open (narrowed). Full chart atlas and transition maps derived and computer-algebra verified (§6.1); inner chart solved in closed form (§6); K_1 entry, K_2 passage (uniform transient bounds, α -correction), and K_3 exit landing (un-blowing to the classical slow manifold) PROVED by invariant-manifold arguments with all computations machine-checked (Lemmas 2–4, §6.2). Remaining: the gluing of the legs (where the uniform Gronwall-level estimates live), time asymptotics.
Kernel-verified certificate layer: finite algebraic obligations of the Theorem 1 certificate and the A.2b/A.3 estimands (Appendix E)	Partial. 16 of 17 staged obligations proved in Lean 4 and kernel-checked, incl. Lemma 3’s completed-square display, both floor inequalities, the exit-time bound, and the A.2b/A.3 estimand identities; the invariant-manifold lemmas and the gluing are untouched by this layer.

Table 1: Status of claims. The fold-repulsion / exit-time theorem and the isolating-block certificate promote the reduced-flow admissibility equivalence to the full singular system at every fixed fold distance. The remaining open items are localized: the joint limit $\delta \rightarrow 0$ at fixed ε , and the cooperative equilibrium neighborhood.

2. Introduction

Capacity-constrained systems appear across domains: scientific publishing pipelines, organisations operating under fixed verification bandwidth, AI deployment where generation outpaces evaluation, and ecological regimes near tipping points [7, 8]. Each features a load variable growing against a bounded capacity, with a coordination cost increasing nonlinearly as load approaches capacity.

This paper is a self-contained mathematical result with stated empirical implications. The theorems and empirical predictions below follow from the local model alone, without dependence on any broader framework. We study a minimal local model in which the fast dynamics realise a smooth saddle-node bifurcation and the slow vector field is singular at the bifurcation boundary. The combination is natural for capacity-constrained mechanisms in which per-unit coordination cost diverges as available slack vanishes, but it lies outside the scope of standard smooth fast-slow theorems [3, 10, 15]. Our contribution is to identify a strict congestion regime, bounded away from both the fold and the slow equilibrium, on which a local Lyapunov certificate for fold-avoidance holds for the full singular system. The certificate is parametrically explicit; its reach is an ε -dependent gap $\delta \gtrsim \varepsilon^{1/4}$ rather than any parameter-fixed minimum distance, and the joint limit $\delta \rightarrow 0$ at fixed ε is left open.

Section 3 verifies the fast bifurcation and the conditional friction-scaling law. Section 4 analyses the reduced slow flow on the attracting branch, formalises the friction barrier as Lyapunov monotonicity on the congestion regime, and states the boundedness equivalence as a reduced-flow result. Section 5 promotes the admissibility equivalence to a fold-repulsion and exit-time theorem for the full singular system on strict congestion subsets, and shows the limitation is an ε -threshold degenerating as $\delta \rightarrow 0$ rather than a parameter-fixed minimum distance. Section 6 addresses the $x = 0$ singularity and presents the formal blow-up heuristic. Sections 7 and 8 state fold-consistent empirical predictions with explicit competing mechanisms.

3. Local Model: Fast Subsystem

3.1 System Definition

Let $x \in \mathbb{R}$ and $V \in \mathbb{R}$. We consider the fast-slow system

$$\varepsilon \dot{x} = f(x, V) := (Q - V) - x^2, \quad (1)$$

$$\dot{V} = g(x) := \alpha - \beta(k/x), \quad (2)$$

with friction coupling $F(x) = k/x$, where $\varepsilon > 0$, $0 < \varepsilon \ll 1$, $Q > 0$, $\alpha > 0$, $\beta > 0$, $k > 0$.

The variable x represents structural slack: the margin between capacity and load. The physical domain is $x > 0$. The fold of the fast subsystem occurs at the boundary $x = 0$.

The fast vector field f is smooth on $\mathbb{R}^2$. The slow vector field g is smooth on $\{x > 0\}$ but singular at $x = 0$: $g(x) \rightarrow -\infty$ as $x \rightarrow 0^+$. This singularity is the central technical feature of the model; we return to it in Section 6.

3.2 Fast-Subsystem Bifurcation

Proposition 1 (*Saddle-node bifurcation of the fast subsystem*). The fast subsystem $\varepsilon \dot{x} = f(x, V)$ defined by (1) satisfies the standard sufficient conditions for a saddle-node (fold) bifurcation at $(x, V) = (0, Q)$.

Proof. Equilibria of the fast subsystem satisfy $f(x, V) = 0$, giving $x = \pm\sqrt{Q - V}$. The attracting branch $x^*(V) = +\sqrt{Q - V}$ has $f_x(x^*, V) = -2x^* < 0$; the repelling branch $x^-(V) = -\sqrt{Q - V}$ has $f_x(x^-, V) > 0$. As $V \rightarrow Q^-$ the branches collide at $(0, Q)$. Verifying the standard saddle-node conditions [1, 2]: $f(0, Q) = 0$,

$f_x(0, Q) = 0$, $f_{xx}(0, Q) = -2 \neq 0$, $f_V(0, Q) = -1 \neq 0$. By the normal-form theorem applied to the smooth fast subsystem, it is locally topologically equivalent to $\varepsilon \dot{u} = \lambda - u^2$ near $(0, Q)$. $\square$

Remark 1. Proposition 1 is a normal-form verification for the smooth fast subsystem. It does not by itself provide a fast-slow fold theorem for the full system (1)–(2), because the slow vector field is singular at the fold point. The full-system result is established for compact subsets of the congestion regime in Section 5.

Proposition 2 (*Conditional friction divergence*). Under the coupling $F(x) = k/x$, the friction along the attracting fast-subsystem equilibrium branch satisfies $F(x^*(V)) \rightarrow +\infty$ as $V \rightarrow Q^-$, with rate $F \sim (Q - V)^{-1/2}$.

Proof. $F(V) = k/x^*(V) = k/\sqrt{Q - V}$, so $\lim_{V \rightarrow Q^-} F(V) = +\infty$ with rate $(Q - V)^{-1/2}$. $\square$

Remark 2 (*Scope of universality and what the empirical protocol tests*). The branch scaling $x^* \sim (Q - V)^{1/2}$ is universal for any saddle-node bifurcation under smooth coordinate changes. The critical-slowness-down observables that follow from this geometry, recovery time τ and stationary variance Var , inherit the universal exponent $-1/2$ independent of the friction coupling form (Section 7). The friction exponent of Proposition 2 itself is model-dependent: $F = k/x^a$ would give $F \sim (Q - V)^{-a/2}$, and the inverse coupling $a = 1$ is a structural assumption when x represents available slack and F represents per-unit coordination cost. The $a = 1$ form has a queueing pedigree: for an M/M/1 server the mean sojourn time is $1/(\text{service} - \text{load})$, so per-unit delay scales as k/slack [20], and the hyperbolic divergence is less arbitrary than a bare postulate. The central hypothesis is qualitative rather than exponent-level: that coordination cost *diverges* as slack vanishes, rather than saturating or diverging only logarithmically. This hyperbolic divergence is what places the model outside the smooth Fenichel and Krupa–Szmolyan theory; the model presumes it rather than adjudicates it, and its direct empirical status is the friction-exponent measurement of Proposition 2, distinct from the CSD protocol. The empirical protocol of Appendix A tests the universal τ and Var exponents, which falsify the saddle-node classification when violated; the friction exponent of Proposition 2 is a separate prediction that would require direct measurement of per-unit coordination cost across the observation window, and is not part of the universal CSD test.

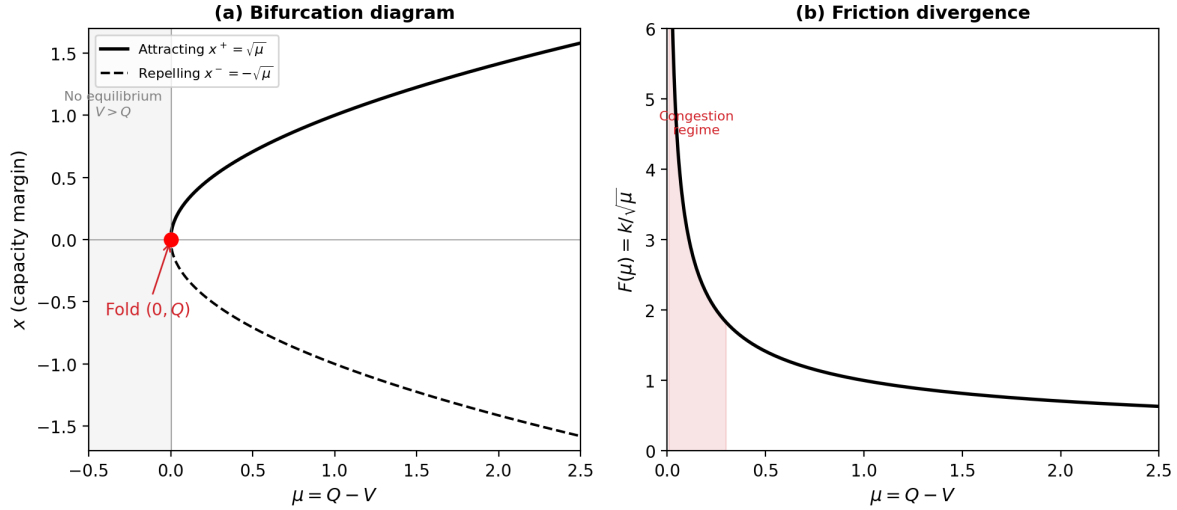


Figure 1. Both panels are plotted against $\mu = Q - V$. (a) Bifurcation diagram of the fast subsystem. The attracting branch $x^*(V) = +\sqrt{Q - V}$ (solid) and the repelling branch $x^-(V) = -\sqrt{Q - V}$ (dashed) collide at the fold $(0, Q)$, i.e., at $\mu = 0$. The left-side shading ($\mu < 0$, equivalently $V > Q$) marks the region where no real equilibrium exists. (b) Friction $F(V) = k/\sqrt{Q - V}$ along the attracting branch, diverging as $V \rightarrow Q^-$ with exponent $-1/2$ in $\mu = Q - V$. The left-side shading (small μ) marks the congestion regime $V_0 < V < Q$ where the attracting branch x^* approaches the fold $x = 0$; the divergence in μ corresponds to a hyperbolic divergence $F = k/x$ in the slack variable along the same branch.

4. Reduced Slow Flow on the Attracting Branch

We analyse the slow equation $\dot{V} = g(V) = \alpha - \beta(k/x)$ along the attracting branch $x = x^*(V) = \sqrt{Q - V}$ as a one-dimensional flow in V , well-defined for $V < Q$.

4.1 Regime Partition

Define $x_0 = \beta k / \alpha$ and $V_0 = Q - (\beta k / \alpha)^2$. The reduced flow $\dot{V} = \alpha - \beta k / x^*(V)$ partitions the attracting branch into:

- **Cooperative regime** ($x^* > x_0$, equivalently $V < V_0$): $\dot{V} > 0$. Load grows because friction is below baseline growth rate α .
- **Congestion regime** ($0 < x^* < x_0$, equivalently $V_0 < V < Q$): $\dot{V} < 0$. Load contracts because friction overwhelms growth.

The reduced flow has a stable fixed point at V_0 .

4.2 Barrier Structure of the Friction

Definition 1 (*Barrier function for fold-avoidance*). On the attracting branch, define

$$L(V) := -F(x^*(V)) = -\frac{k}{\sqrt{Q - V}}.$$

L is smooth and strictly decreasing in V on $(-\infty, Q)$, with $L(V) \rightarrow -\infty$ as $V \rightarrow Q^-$. The certificate W assembled from L below is a barrier certificate in the sense of [17], the object the control-barrier-function literature [18] uses for safety-critical control.

Proposition 3 (*Barrier monotonicity on the congestion regime*). On the congestion regime $\{V_0 < V < Q\}$, the reduced slow flow satisfies $\dot{V} < 0$, and consequently

$$\frac{dL}{dt} > 0$$

along trajectories of the reduced flow. Equivalently, F is strictly decreasing along reduced trajectories in the congestion regime: the trajectory moves away from the fold.

Proof. Differentiating along the reduced flow:

$$\frac{dL}{dt} = -\frac{d}{dt} \left(\frac{k}{\sqrt{Q - V}} \right) = -\frac{k\dot{V}}{2(Q - V)^{3/2}}.$$

For $V < Q$, the denominator is positive, so dL/dt has the sign of $-\dot{V}$. On the congestion regime, $\sqrt{Q - V} < \beta k / \alpha$ gives $\dot{V} = \alpha - \beta k / \sqrt{Q - V} < 0$, hence $dL/dt > 0$. $\square$

Remark 3. The barrier monotonicity is active in the congestion regime, where the trajectory is close to the fold and the barrier drives it back. In the cooperative regime, $\dot{V} > 0$ and the dynamics are growth-dominated; the trajectory may approach the boundary $V = V_0$ from below.

4.3 Structural Admissibility Equivalence

Criterion 1 (*Boundedness equivalence*). On the reduced slow flow restricted to the attracting branch, the condition

$$\exists M < \infty \text{ such that } F(x^*(V(t))) \leq M \text{ for all } t \geq 0$$

is equivalent to the trajectory remaining bounded away from the fold:

$\exists \delta > 0$ such that $x^*(V(t)) \geq \delta$ for all $t \geq 0$.

Proof. If $F(x^*(V)) = k/\sqrt{Q-V} \leq M$, then $x^*(V) \geq k/M$; take $\delta = k/M$. Conversely if $x^*(V) \geq \delta$, then $F(x^*(V)) \leq k/\delta$; take $M = k/\delta$. $\square$

Remark 4. Criterion 1 is a structural equivalence on the reduced flow. The promotion to a fold-repulsion and exit-time statement for the full two-dimensional singular system is given in Section 5, restricted to strict congestion subsets bounded away from both the fold and the cooperative equilibrium.

5. Fold Repulsion in the Full System

We promote the reduced-flow admissibility equivalence to a fold-repulsion and exit-time statement for the full singular system on a compact subset of the congestion regime, bounded away from both the fold and the slow equilibrium. The result is a local Lyapunov-decay certificate for fold-avoidance in the full two-dimensional phase plane.

5.1 Setup

We work in fast time t , with the system

$$\dot{x} = (Q - V) - x^2, \quad \dot{V} = \varepsilon g(x), \quad g(x) := \alpha - \frac{\beta k}{x}.$$

Let $x_0 = \beta k/\alpha$ and $V_0 = Q - x_0^2$. For $r > 0$ and $\delta \in (0, x_0)$, define the strict congestion interval

$$J(r, \delta) := [V_0 + r, Q - \delta^2],$$

assumed nonempty (i.e., $V_0 + r < Q - \delta^2$, equivalently $\delta^2 < x_0^2 - r$). The slow manifold over J is $x = x^*(V) = \sqrt{Q - V}$, with $x^*(V) \in [\delta, \sqrt{x_0^2 - r}]$.

For constants $C > 0$ and $\varepsilon > 0$ to be specified, define the **slow-manifold tube**

$$T_{C,\varepsilon}(r, \delta) := \{(x, V) : V \in J(r, \delta), |x - x^*(V)| \leq C\sqrt{\varepsilon}\}.$$

The deviation variable is $y := x - x^*(V)$.

What the certificate must control. The nontriviality is not that the reduced slow flow avoids the fold: on the manifold it flows toward V_0 , away from $x = 0$. It is that the full-system fast-fiber dynamics, coupled to the slow direction through the diverging $g(x)$, do not open an escape route back toward $x = 0$. Term I supplies the fast restoration, the fast-fiber decay $-y^2(x + x^*)/\varepsilon$ that dominates the residual III_{res} ; the weight w then makes the weighted barrier (Term II) dominate the worst-case cross-coupling (Term IV, through the tube-wide constant $\Gamma_b = 2\beta k/\delta - \alpha$). Making the singularity's dynamical load explicit is what separates this from a smooth Fenichel reduction.

5.2 The Lyapunov Function

Define

$$W(x, V) := w \cdot \frac{k}{\sqrt{Q - V}} + \frac{y^2}{2\varepsilon} + b(V)y, \quad b(V) := \frac{g(x^*(V))}{4(Q - V)},$$

with a constant weight $w > 0$ fixed in the proof. The first term is the reduced-flow barrier $-L(V)$ of Definition 1, scaled by w , penalizing approach to the fold; the weight is the construction's key degree of

freedom (Remark 5). The second term is a fast-fiber penalty at the Fenichel scaling. The third term is a linear-in- y correction whose role is to eliminate the leading-order cross coupling between the fast and slow directions; it does not involve w . The function W is proper and bounded below on the tube: completing the square, $W = \frac{1}{2\varepsilon}(y + \varepsilon b)^2 + wk/\sqrt{Q - V} - \frac{\varepsilon}{2}b(V)^2$, a sum of a nonnegative fast-fiber term, the positive barrier $wk/\sqrt{Q - V}$, and a benign $O(\varepsilon)$ shift, so its sublevel sets within the tube are compact, and on them the decay estimate is a Lyapunov-decay certificate.

5.3 Structural Parameters

The proof uses the following ε -free constants depending on $(r, \delta, \alpha, \beta, k)$:

- **Congestion gap on the slow manifold.** For $V \in J$, $x^*(V) \leq \sqrt{x_0^2 - r} < x_0$, so $g(x^*(V)) \leq -\gamma_0$ with

$$\gamma_0(r) := \frac{\beta k}{\sqrt{x_0^2 - r}} - \alpha > 0.$$

- **Fast restoration.** For $(x, V) \in T_{C, \varepsilon}$ with $C\sqrt{\varepsilon} \leq \delta/2$, $x + x^*(V) \geq 2\delta - C\sqrt{\varepsilon} \geq \delta$.
- **Cross-coupling bounds.** On the tube, $g(x) < 0$ with $|g(x)| = \beta k/x - \alpha$, provided $x < x_0$ throughout the tube (this is ensured by the constraint on ε stated below). For $C\sqrt{\varepsilon} \leq \delta/2$, $x \geq \delta/2$, giving the tube-wide bound

$$|g(x)| \leq \Gamma_b(\delta) := \frac{2\beta k}{\delta} - \alpha;$$

on the slow manifold over J , where $x^*(V) \geq \delta$, the sharper bound

$$|g(x^*)| \leq \Gamma_\sharp(\delta) := \frac{\beta k}{\delta} - \alpha$$

holds. Both are used in Term IV; no $\varepsilon \rightarrow 0$ limit is invoked anywhere in the proof.

- **Slow repulsion lower bound.** $(Q - V)^{-3/2} \geq (x_0^2 - r)^{-3/2} =: \mu_0$ for $V \in J$.

5.4 Main Result

Theorem 1 (*Lyapunov decay and fold-repulsion funnel; explicit constants*). Fix $r > 0$ and $\delta \in (0, \sqrt{x_0^2 - r})$ (equivalently $V_0 + r < Q - \delta^2$, so $J(r, \delta)$ is nonempty), and fix the tube half-width $C > 0$. Write $u_+ := \sqrt{x_0^2 - r}$ and set the slow-term weight

$$w := \frac{\Gamma_\sharp(\delta) \Gamma_b(\delta)}{k \gamma_0(r)},$$

the rate constants

$$\kappa_0 := \frac{\mu_0(r) \Gamma_\sharp(\delta) \Gamma_b(\delta)}{8}, \quad \kappa := \frac{\kappa_0}{2},$$

and the floor and margin constants

$$L_- := \frac{wk}{u_+}, \quad c_M := \frac{\gamma_0(r) C}{4 u_+^2}, \quad c_\delta := \frac{\Gamma_\sharp(\delta)^2}{32 \delta^4}, \quad c_T := \frac{\gamma_0(r)^2}{32 u_+^4}.$$

Let $\varepsilon_0 := \min\{\varepsilon_{\text{tube}}, \varepsilon_{\text{res}}, \varepsilon_{\text{Lip}}, \varepsilon_1, \varepsilon_x, \varepsilon_{\text{win}}\}$ be the six-term smallness threshold whose entries — each a closed form in $(\alpha, \beta k, r, \delta, C)$ — are displayed in Appendix D and motivated, one bound per entry, in the Assembly step of the proof.

Then for all $\varepsilon \in (0, \varepsilon_0]$,

$$\frac{dW}{dt} \leq -\kappa \varepsilon \quad \text{on } T_{C,\varepsilon}(r, \delta).$$

For any M in the window

$$L_- + c_M \sqrt{\varepsilon} < M < \min \left\{ L_- + \frac{C^2}{2} - c_M \sqrt{\varepsilon}, \frac{wk}{\delta} - c_\delta \varepsilon \right\}$$

(nonempty when the cap is the first entry; when the cap is the second entry, assume additionally the explicit fold-gap condition $c_M \sqrt{\varepsilon} + c_\delta \varepsilon < wk(1/\delta - 1/u_+)$ — Appendix D shows the first case needs no extra condition and gives a sufficient closed form for the second), the funnel $\{W \leq M\} \cap T_{C,\varepsilon}$ is nonempty, and every trajectory initialized in it has W strictly decreasing at rate $\kappa \varepsilon$; it therefore never reaches the fast-fiber boundaries $\{|y| = C\sqrt{\varepsilon}\}$ or the fold-side boundary $\{V = Q - \delta^2\}$, stays in the tube with $x \geq \delta/2$, and exits through the cooperative-side face $\{V = V_0 + r\}$ in time at most

$$T \leq \frac{M - L_- + c_T \varepsilon}{\kappa \varepsilon} \leq \frac{C^2/2 + c_T \varepsilon}{\kappa \varepsilon},$$

the second inequality because the window cap forces $M - L_- < C^2/2$.

The funnel is not forward-invariant: the fifth threshold gives $\dot{V} = \varepsilon g(x) < 0$ throughout $T_{C,\varepsilon}$, so every trajectory leaves in finite time (Proposition 5). This is a Lyapunov-decay and exit-time certificate; invariance can hold only in the cooperative-equilibrium region of Remarks 4a and 9, which this theorem does not reach. Every constant above is closed-form: the structural constants are collected in Table 2, the threshold entries are displayed in Appendix D, and they match the Appendix C computation after substituting the chosen C (Appendix C assembles ε_0 and verifies the certified run at $C = 1$, and runs its wider demonstration tube at $C = 2$). The statement contains no asymptotic notation.

Proof architecture. The dependency structure of the lemma chain is

$$\begin{aligned} \text{tube geometry (§5.1, §5.3)} &\Rightarrow \text{bounds on } g, g', b, b' \Rightarrow \begin{cases} \text{I}/2 + \text{III}_{\text{res}} \leq 0 & (\text{L1-L3}) \\ \text{II} + \text{IV} + \text{V} \leq -\kappa_0 \varepsilon / 2 & (\text{L4-L5}) \end{cases} \\ &\Rightarrow \frac{dW}{dt} \leq -\kappa \varepsilon \Rightarrow \text{floors and window (L6, Lemma 1)} \Rightarrow \text{rated exit (L7)}. \end{aligned}$$

Proof (as a lemma chain; each step is a self-contained, machine-checkable claim, and each cites only earlier steps or the displayed constants of §5.3).

L1 (derivative identity). Compute $dW/dt = (\partial_x W)\dot{x} + (\partial_V W)\dot{V}$ with

$$\partial_x W = \frac{y}{\varepsilon} + b(V), \quad \partial_V W = \frac{wk}{2(Q-V)^{3/2}} + \frac{y}{2\varepsilon\sqrt{Q-V}} + b'(V)y + \frac{b(V)}{2\sqrt{Q-V}},$$

and the dynamics $\dot{x} = -y(x + x^*)$ (using $(Q - V) - x^2 = -(x - x^*)(x + x^*)$) and $\dot{V} = \varepsilon g(x)$.

Substituting and grouping, with $u := \sqrt{Q - V}$:

$$\frac{dW}{dt} = \underbrace{-\frac{y^2(x + x^*)}{\varepsilon}}_{\text{I}} + \underbrace{\frac{\varepsilon wk g(x)}{2(Q - V)^{3/2}}}_{\text{II}} + \underbrace{\left[\frac{g(x)}{2u} - b(V)(x + x^*) \right] y}_{\text{III}_{\text{lin}}} + \underbrace{\frac{\varepsilon b(V) g(x)}{2u}}_{\text{IV}} + \underbrace{\varepsilon b'(V) g(x) y}_{\text{V}}.$$

Only Term II carries the weight w ; the cancellation below and Terms I, III, V are w -free.

L2 (linear cross-term cancellation). With $b(V) = g(x^*)/[4(Q - V)]$, write $x = x^* + y$ so $x + x^* = 2u + y$, and use $g(x^* + y) - g(x^*) = \beta k y/[u(u + y)]$:

$$\frac{g(x)}{2u} - \frac{g(x^*)(2u + y)}{4u^2} = \frac{2u[g(x^* + y) - g(x^*)] - y g(x^*)}{4u^2} = \frac{y}{4u^2} \left[\frac{2\beta k}{u + y} - g(x^*) \right].$$

Hence

$$\text{III}_{\text{lin}} = \frac{y^2}{4u^2} \left[\frac{2\beta k}{u + y} - g(x^*) \right] =: \text{III}_{\text{res}}.$$

The linear-in- y term has been eliminated; the residual is quadratic.

L3 (residual domination). On the tube, $u \geq \delta$ and $u + y \geq \delta/2$ (for $C\sqrt{\varepsilon} \leq \delta/2$). On congestion, $-g(x^*) \leq \Gamma_{\sharp}(\delta) = \beta k/\delta - \alpha < \beta k/\delta$, combined with $2\beta k/(u + y) \leq 4\beta k/\delta$:

$$|\text{III}_{\text{res}}| \leq \frac{y^2}{4\delta^2} \left[\frac{4\beta k}{\delta} + \Gamma_{\sharp} \right] \leq \frac{5\beta k y^2}{4\delta^3}.$$

Domination by Term I. $|\text{I}| \geq \delta y^2/\varepsilon$. For $\varepsilon \leq 2\delta^4/(5\beta k)$:

$$\frac{\text{I}}{2} + \text{III}_{\text{res}} \leq -\frac{\delta y^2}{2\varepsilon} + \frac{5\beta k y^2}{4\delta^3} \leq 0,$$

leaving $\text{I}/2 \leq -\delta y^2/(2\varepsilon)$ available.

L4 (slow-term closure). *Bounding Term IV.* On the tube, $|g(x^*)| \leq \Gamma_{\sharp}$ (the manifold bound, since $x^*(V) \geq \delta$ on J) and $|g(x)| \leq \Gamma_{\flat}$ (the tube-wide bound, since $x \geq \delta/2$; both negative subject to $x < x_0$ throughout the tube, see the ε constraint below). Hence

$$|\text{IV}| = \frac{\varepsilon |g(x^*)| |g(x)|}{8(Q - V)^{3/2}} \leq \frac{\varepsilon \Gamma_{\sharp} \Gamma_{\flat}}{8(Q - V)^{3/2}}.$$

This bound holds at finite ε ; no limit is taken.

Negativity of Term II. g is Lipschitz on $[\delta/2, 2x_0]$ with $|g'(x)| = \beta k/x^2 \leq 4\beta k/\delta^2 =: L_g$. For ε small enough that $L_g C\sqrt{\varepsilon} \leq \gamma_0/2$, $g(x) \leq -\gamma_0/2$ on the tube. Hence

$$\text{II} \leq -\frac{\varepsilon w k \gamma_0}{4(Q - V)^{3/2}}.$$

Combining II and IV.

$$\text{II} + \text{IV} \leq \frac{\varepsilon}{(Q - V)^{3/2}} \left[-\frac{w k \gamma_0}{4} + \frac{\Gamma_{\sharp} \Gamma_{\flat}}{8} \right].$$

By the choice $w = \Gamma_{\sharp} \Gamma_{\flat}/(k \gamma_0)$ in the theorem statement,

$$-\frac{w k \gamma_0}{4} + \frac{\Gamma_{\sharp} \Gamma_{\flat}}{8} \leq -\frac{\Gamma_{\sharp} \Gamma_{\flat}}{8} < 0$$

for every $\delta \in (0, \sqrt{x_0^2 - r})$: no further condition on δ arises. At the pinned weight this gives $\text{II} + \text{IV} \leq -\varepsilon \mu_0 \Gamma_{\sharp} \Gamma_{\flat}/8$ directly, so set $\kappa_0 := \mu_0 \Gamma_{\sharp} \Gamma_{\flat}/8 > 0$, depending only on $(\alpha, \beta k, \delta, r)$ (split-invariant, Remark 5), so that $\text{II} + \text{IV} \leq -\kappa_0 \varepsilon$ on the tube.

L5 (Term V domination). Compute

$$b'(V) = -\frac{\beta k}{8(Q-V)^{5/2}} + \frac{g(x^*)}{4(Q-V)^2},$$

so, writing $u := \sqrt{Q-V} \geq \delta$ for the fiber's fold distance and using the manifold value $|g(x^*(V))| = \beta k/u - \alpha \leq \beta k/u$, $|b'(V)| \leq \beta k/(8u^5) + |g(x^*(V))|/(4u^4) \leq \beta k/(8u^5) + \beta k/(4u^5) = 3\beta k/(8u^5)$ pointwise, and $|g(x)| \leq 2\beta k/u$ for the tube state (constraint 1 gives $x \geq u - C\sqrt{\varepsilon} \geq u/2$). Hence, fiberwise,

$$|V| \leq \varepsilon |b'(V)| |g(x)| |y| \leq \frac{3C(\beta k)^2}{4} \frac{\varepsilon^{3/2}}{u^6}.$$

The proof already establishes $\text{II} + \text{IV} \leq -\varepsilon \Gamma_{\sharp} \Gamma_b / (8u^3)$ pointwise (before $\mu_0 \leq u^{-3}$ is applied). Requiring $|V| \leq |\text{II} + \text{IV}|/2$ fiberwise gives $\varepsilon^{1/2} \leq \Gamma_{\sharp} \Gamma_b u^3 / (12C(\beta k)^2)$, worst at $u = \delta$; hence, once $\varepsilon \leq \varepsilon_1 := (\Gamma_{\sharp} \Gamma_b \delta^3 / (12C(\beta k)^2))^2$, we have $|V| \leq |\text{II} + \text{IV}|/2$ fiberwise, so $\text{II} + \text{IV} + \text{V} \leq (\text{II} + \text{IV})/2 \leq -\kappa_0 \varepsilon / 2$. Since $\varepsilon_1 \sim \delta^2$, Term V is dominated by the δ^4 constraints below and does not bind the certified reach.

Assembly. The bounds above require precisely the six named thresholds of the statement (closed forms in Appendix D):

- $C\sqrt{\varepsilon} \leq \delta/2$ ($\varepsilon \leq \varepsilon_{\text{tube}} = \delta^2/(4C^2)$), so that the tube stays in $\{x \geq \delta/2\}$ and the fast-restoration / residual bounds apply.
- $\varepsilon \leq \varepsilon_{\text{res}} = 2\delta^4/(5\beta k)$, so that Term I dominates the residual (L3).
- $L_g C\sqrt{\varepsilon} \leq \gamma_0/2$ ($\varepsilon \leq \varepsilon_{\text{Lip}} = (\gamma_0/(2L_g C))^2$), so that Term II is negative (L4).
- $\varepsilon \leq \varepsilon_1 = (\Gamma_{\sharp} \Gamma_b \delta^3 / (12C(\beta k)^2))^2 \sim \delta^2$, so that Term V is subdominant (L5; dominated by the δ^4 constraints above).
- $C\sqrt{\varepsilon} \leq x_0 - u_+$, equivalently $\varepsilon \leq \varepsilon_x = (r/(C(x_0 + u_+)))^2$, so that $x < x_0$ throughout the tube and the bounds on $|g(x)|$ and $|g(x^*)|$ in terms of $\Gamma_{\sharp}$ remain valid. (Without this, the tube can cross into $\{x \geq x_0\}$ at the upper- V end of J , where $g(x) \geq 0$ and the closure condition no longer applies.)
- $\varepsilon \leq \varepsilon_{\text{win}}(\delta, r)$, so that the M -window is nonempty and the boundary minimum of W sits on the cooperative face (Lemma 1, used by L6–L7).

For $\varepsilon \in (0, \varepsilon_0]$:

$$\frac{dW}{dt} \leq \underbrace{\frac{1}{2}}_{\leq 0} + \underbrace{(\text{II} + \text{IV} + \text{V})}_{\leq -\kappa_0 \varepsilon / 2} + \underbrace{(\text{I}/2 + \text{III}_{\text{res}})}_{\leq 0} \leq -\frac{\kappa_0 \varepsilon}{2}.$$

With $\kappa := \kappa_0/2$ this is the claimed decay bound; the funnel, window, and exit-time claims are steps L6–L7 below, stated after the floor lemma they rely on.

Proposition 4 (*Local asymptotic stability of the cooperative equilibrium*). The point $(x_0, V_0) = (\beta k/\alpha, Q - (\beta k/\alpha)^2)$ is an asymptotically stable equilibrium of (1)–(2) for every $\varepsilon > 0$.

Proof. Linearisation around (x_0, V_0) with $u = x - x_0$, $\eta = V - V_0$ gives, in fast time,

$$\dot{u} = -2x_0 u - \eta + O(u^2, u\eta), \quad \dot{\eta} = \frac{\varepsilon \alpha}{x_0} u + O(\varepsilon u^2).$$

The Jacobian is $J(\varepsilon) = \begin{pmatrix} -2x_0 & -1 \\ \varepsilon \alpha / x_0 & 0 \end{pmatrix}$, with $\text{tr } J = -2x_0 < 0$ and $\det J = \varepsilon \alpha / x_0 > 0$ for every $\varepsilon > 0$. Both eigenvalues have strictly negative real part (Hurwitz). By Lyapunov's indirect method, (x_0, V_0) is locally asymptotically stable. The eigenvalues separate as $\varepsilon \rightarrow 0$: $\lambda_{\text{fast}} \approx -2x_0$ and $\lambda_{\text{slow}} \approx -\varepsilon \alpha / (2x_0^2)$, exhibiting the standard fast–slow time-scale separation. $\square$

Remark 4a (*Matching to Theorem 1*). Proposition 4 supplies a certified neighborhood $N(\varepsilon)$ around (x_0, V_0) ; Theorem 1 supplies a certified region $T_{C,\varepsilon}(r, \delta)$ on $V \in [V_0 + r, Q - \delta^2]$. The neighborhood $N(\varepsilon)$ is forward-invariant under the autonomous flow via the quadratic Lyapunov function of the Hurwitz linearization at the cooperative equilibrium (x_0, V_0) (Proposition 4); on $T_{C,\varepsilon}(r, \delta)$, by contrast, $\dot{V} < 0$ forces finite-time exit, so there Theorem 1 gives a fold-repulsion funnel rather than invariance. Whether $N(\varepsilon)$ can be chosen large enough to overlap $T_{C,\varepsilon}(r, \delta)$ at the lower- V boundary $V = V_0 + r$ depends on the quantitative Lyapunov-function scaling at the cooperative equilibrium, which decreases in the slow direction as $\varepsilon \rightarrow 0$. Because the attracting branch is normally hyperbolic near the cooperative equilibrium, where g is smooth, Fenichel's theorem [3] supplies the qualitative matching; only the quantitative overlap range is left as the open item in the Status table.

Summary of structural constants. The proof above uses the ε -free constants depending on $(r, \delta, \alpha, \beta, k)$, the rate constants κ_0 and κ , and the sufficiency threshold ε_0 . They are summarised in Table 2.

Constant	Definition	Role
$\gamma_0(r)$	$\beta k / \sqrt{x_0^2 - r} - \alpha$	Lower bound on $-g(x^*(V))$ over $V \in J$; congestion gap on the slow manifold.
$\Gamma_\#(\delta)$	$\beta k / \delta - \alpha$	Upper bound on $ g(x^*) $ on the slow manifold over J .
$\Gamma_b(\delta)$	$2\beta k / \delta - \alpha$	Upper bound on $ g(x) $ on the tube; worst-case cross-coupling magnitude at finite ε .
$\mu_0(r)$	$(x_0^2 - r)^{-3/2}$	Lower bound on $(Q - V)^{-3/2}$ for $V \in J$; slow-repulsion strength.
$L_g(\delta)$	$4\beta k / \delta^2$	Lipschitz constant for g on the tube.
$w(\delta, r)$	$\Gamma_\# \Gamma_b / (k\gamma_0)$	Slow-term weight closing II + IV for every δ (see Remark 5).
κ_0	$\mu_0 \Gamma_\# \Gamma_b / 8$	Tube-wide slow-decay constant (II + IV closure; Remark 5).
κ	$\kappa_0 / 2$	Net decay rate $dW/dt \leq -\kappa\varepsilon$; sets the exit-time bound.
L_-	$wk / u_+, u_+ = \sqrt{x_0^2 - r}$	Cooperative-face barrier level; base of the M -window.
c_b	$\Gamma_\# C / (4\delta^2)$	Tube-wide bound $ by \leq c_b \sqrt{\varepsilon}$ on the cross term.
c_M	$\gamma_0 C / (4u_+^2)$	M -window margin; fiber-floor correction at u_+ (L6–L7).
c_δ	$\Gamma_\#^2 / (32\delta^4)$	Fold-side floor correction $wk/\delta - c_\delta\varepsilon$ (L6).
c_T	$\gamma_0^2 / (32u_+^4)$	Exit-time numerator margin $M - L_- + c_T\varepsilon$ (L7).

Table 2. Structural constants in the proof of Theorem 1.

L6 (boundary floor). Completing the square (§5.2),

$$W = \frac{(y + \varepsilon b)^2}{2\varepsilon} + h(u), \quad h(u) := \frac{wk}{u} - \frac{\varepsilon}{2} b(V)^2,$$

so $W \geq h(u)$ pointwise on every slow fiber. The three boundary faces of $T_{C,\varepsilon}(r, \delta)$ then carry the floors:

- *Fold side* $\{u = \delta\}$: here $g(\delta) = -\Gamma_\#(\delta)$, so $|b| = \Gamma_\#(\delta)/(4\delta^2)$ and $W \geq h(\delta) = wk/\delta - c_\delta\varepsilon$.

- *Fast fibers* $\{|y| = C\sqrt{\varepsilon}\}$: $W = wk/u + C^2/2 \pm bC\sqrt{\varepsilon} \geq \varphi(u) := wk/u + C^2/2 - |b|C\sqrt{\varepsilon}$. By Lemma 1 (below; $\varepsilon \leq \varepsilon_{\text{win}}$) the envelope is minimized at u_+ , where $g(u_+) = -\gamma_0(r)$ and $|b| = \gamma_0/(4u_+^2)$, giving $W \geq L_- + C^2/2 - c_M\sqrt{\varepsilon}$.
- *Cooperative side* $\{u = u_+\}$: $W \geq h(u_+) = L_- - c_T\varepsilon$.

Tube-wide the cross term obeys $|by| \leq c_b\sqrt{\varepsilon}$ with $c_b := \Gamma_\#(\delta)C/(4\delta^2)$; on the cooperative face the sharper $c_M\sqrt{\varepsilon}$ applies. The interior minimum of W over the closed tube equals $h(u_+) = L_- - c_T\varepsilon$, attained at $(u, y) = (u_+, -\varepsilon b(u_+))$, admissible because $\varepsilon \leq \varepsilon_{\text{win}}$ forces $\varepsilon\gamma_0/(4u_+^2) \leq C\sqrt{\varepsilon}$.

L7 (exit-time assembly). Let M lie in the window of the statement, i.e. strictly below both non-exit floors of L6. Along a trajectory initialized in $\{W \leq M\} \cap T_{C,\varepsilon}$ the decay bound gives $dW/dt \leq -\kappa\varepsilon < 0$, so $W \leq M$ persists; the trajectory can therefore reach neither the fast-fiber faces ($W \geq L_- + C^2/2 - c_M\sqrt{\varepsilon} > M$ there) nor the fold-side face ($W \geq wk/\delta - c_\delta\varepsilon > M$ there), and satisfies $x \geq \delta - C\sqrt{\varepsilon} \geq \delta/2$ until it exits. Since $\dot{V} = \varepsilon g(x) < 0$ throughout the tube (Proposition 5), it exits in finite time T through the cooperative face, where $W(T) \geq L_- - c_T\varepsilon$ (L6). Integrating the decay, $W(T) \leq M - \kappa\varepsilon T$; combining,

$$T \leq \frac{M - L_- + c_T\varepsilon}{\kappa\varepsilon}.$$

The funnel is nonempty: the point $(u, y) = (u_+, 0)$ has $W = L_- < M$. Window nonemptiness is Lemma 1 plus, in the fold-cap case, the explicit fold-gap condition assumed in the statement. This is a decay-and-exit certificate rather than an invariance statement. $\square$

The floor location and window nonemptiness are secured by the following lemma, whose constants enter ε_{win} .

Lemma 1 (Floor location and window nonemptiness). With w pinned and $u_+ := \sqrt{x_0^2 - r}$, if $\varepsilon \leq \varepsilon_{\text{win}}(\delta, r)$ then the boundary minimum of W over $\partial T_{C,\varepsilon}$ is attained on the cooperative face $u = u_+$, and the M -window is nonempty.

Proof. With $b = g(x^*)/(4u^2)$ and $|g(x^*)| = \beta k/u - \alpha \leq \beta k/u$, $|(b^2)'| = |g(x^*)|(\beta k/u + 2|g(x^*)|)/(8u^5) \leq 3(\beta k)^2/(8u^7)$ and $|d|b|/du| = (\beta k/u + 2|g(x^*)|)/(4u^3) \leq 3\beta k/(4u^4)$. The interior floor is $h(u) := wk/u - (\varepsilon/2)b^2$, nonincreasing once $(\varepsilon/2) \cdot 3(\beta k)^2/(8u^7) \leq wk/u^2$, i.e. $\varepsilon \leq 16wku^5/(3(\beta k)^2)$, worst at $u = \delta$ (first entry of ε_{win}). The fiber-face floor is the envelope $\varphi(u) := wk/u + C^2/2 - |b(u)|C\sqrt{\varepsilon}$, and φ is nonincreasing once $C\sqrt{\varepsilon} \cdot 3\beta k/(4u^4) \leq wk/u^2$, i.e. $\varepsilon \leq (4wku^2/(3\beta kC))^2$, worst at $u = \delta$ (second entry). Both floors are therefore minimized at u_+ , where $|g(x^*)| = \gamma_0$, $|b| \leq \gamma_0/(4u_+^2)$, and $(\varepsilon/2)b^2 \leq \varepsilon\gamma_0^2/(32u_+^4)$; so the boundary minimum of W sits on the cooperative face. For nonemptiness, when $wk(1/\delta - 1/u_+) \geq C^2/2$ the cap is $L_- + C^2/2$ and the fiber-floor correction stays below the $C^2/2$ gap once $\gamma_0 C\sqrt{\varepsilon}/(4u_+^2) \leq C^2/4$, i.e. $\varepsilon \leq C^2u_+^4/\gamma_0^2$ (third entry); when $wk(1/\delta - 1/u_+) < C^2/2$ the cap is wk/δ and the window width is the fold gap $wk(1/\delta - 1/u_+)$ itself, positive by the same h -monotonicity ($h(\delta) - h(u_+)$ equals the gap minus $O(\varepsilon)$). At the pinned weight the three ε_{win} entries scale as δ^3 , δ^0 , δ^0 (the last two r -degenerate), each weaker than the δ^4 assembly constraints, so the reach $\delta \gtrsim \varepsilon^{1/4}$ is untouched. $\square$

Remark (explicit fold-gap margin). In the second (fold-cap) case of Lemma 1's nonemptiness argument, Theorem 1 carries the explicit condition $c_M\sqrt{\varepsilon} + c_\delta\varepsilon < wk(1/\delta - 1/u_+)$ as a stated hypothesis: for δ near u_+ the fold gap is small, and this is a genuine additional smallness requirement on ε that the six-term ε_0 does not enforce. The sufficient closed form given in the statement makes Lemma 1's "gap minus $O(\varepsilon)$ " explicit rather than leaving it asymptotic.

5.5 Isolating-Block Certificate for the Whole Tube

Theorem 1's funnel consists of the sublevel sets $\{W \leq M\}$ with M below the fast-fiber and fold-side floors of W ; initial conditions near the fold-side face or the fast-fiber boundary lie above those floors and belong to no such set. The following face-flux argument closes that gap: it certifies *every* initial condition in the tube, at the cost of giving no decay rate.

Proposition 5 (Exit only through the cooperative-side face). Fix $r > 0$, $\delta \in (0, \sqrt{x_0^2 - r})$, and $C > 0$. Suppose

$$\varepsilon < \varepsilon_{\text{blk}} := \min \left\{ \frac{\delta^2}{4C^2}, \left(\frac{2C\delta^2}{\Gamma_b(\delta)} \right)^2, \left(\frac{r}{C(x_0 + \sqrt{x_0^2 - r})} \right)^2 \right\}.$$

Then on $T_{C,\varepsilon}(r, \delta)$: the fold-side face $\{V = Q - \delta^2\}$ and both fast-fiber faces $\{y = \pm C\sqrt{\varepsilon}\}$ are strictly inflowing, and $\dot{V} < 0$ holds uniformly. Consequently every trajectory starting in $T_{C,\varepsilon}$ remains in it, with $x \geq \delta/2$, until it exits through the cooperative-side face $\{V = V_0 + r\}$ in time at most

$$T_{\text{exit}} \leq \frac{V_{\text{init}} - V_0 - r}{\varepsilon c_5}, \quad c_5 := \frac{\beta k}{x_{\text{max}}} - \alpha > 0, \quad x_{\text{max}} := \sqrt{x_0^2 - r} + C\sqrt{\varepsilon}$$

($c_5 > 0$ precisely because the third entry of ε_{blk} gives $C\sqrt{\varepsilon} < x_0 - \sqrt{x_0^2 - r}$, i.e. $x_{\text{max}} < x_0$). No trajectory starting in the tube approaches the fold before it exits; global non-return is not certified here (Remark 4b).

Proof. The deviation variable satisfies $\dot{y} = \dot{x} - (x^*)'(V) \dot{V} = -y(x + x^*) + \varepsilon g(x)/(2u)$. The third constraint keeps the tube inside $\{x < x_0\}$, so $g(x) < 0$ with $|g(x)| \leq \Gamma_b$ (first constraint: $C\sqrt{\varepsilon} \leq \delta/2$, so $x \geq \delta/2$). On the upper fiber face $y = +C\sqrt{\varepsilon}$: $\dot{y} = -C\sqrt{\varepsilon}(x + x^*) + \varepsilon g(x)/(2u) < 0$, both terms negative, so inflowing. On the lower fiber face $y = -C\sqrt{\varepsilon}$, using $x + x^* \geq \delta$ and $u \geq \delta$:

$$\dot{y} \geq C\sqrt{\varepsilon}\delta - \frac{\varepsilon\Gamma_b}{2\delta} > 0$$

by the second constraint ($\sqrt{\varepsilon} < 2C\delta^2/\Gamma_b$) so inflowing. On the fold-side face $\{V = Q - \delta^2\}$: $\dot{V} = \varepsilon g(x) < 0$, so inflowing. Finally, on the tube $x \leq x_{\text{max}} = \sqrt{x_0^2 - r} + C\sqrt{\varepsilon} < x_0$ (third constraint), and g is increasing, so $\dot{V} = \varepsilon g(x) \leq \varepsilon g(x_{\text{max}}) = -\varepsilon c_5 < 0$ uniformly; V therefore decreases at rate at least εc_5 from V_{init} to $V_0 + r$, which is the displayed exit-time bound. The naive manifold bound $-g(x) \geq \gamma_0(r)$ is *not* valid on the tube (the tube sits above the manifold by up to $C\sqrt{\varepsilon}$); c_5 is the correct tube-wide constant. The only non-inflowing face is the cooperative-side face, and while in the tube $x \geq x^*(V) - C\sqrt{\varepsilon} \geq \delta/2$. $\square$

Corollary 1 (*Rated exit for all tube initial conditions*). For $\varepsilon \leq \varepsilon_0(\delta, r)$ and $\varepsilon < \varepsilon_{\text{blk}}$, every trajectory starting in $T_{C,\varepsilon}(r, \delta)$ stays in the tube with $x \geq \delta/2$ and exits through the cooperative-side face $\{V = V_0 + r\}$, without fold approach, in time at most

$$\frac{\sup_T W - \inf_T W}{\kappa\varepsilon} \leq \frac{wk/\delta - L_- + C^2/2 + c_b\sqrt{\varepsilon} + c_T\varepsilon}{\kappa\varepsilon};$$

at the pinned weight the leading term is exact: $(wk/\delta)/(\kappa\varepsilon) = 16/(\gamma_0\mu_0\delta\varepsilon)$.

Proof. Theorem 1 gives $dW/dt \leq -\kappa\varepsilon$ at *every* point of $T_{C,\varepsilon}$, so W decreases at rate at least $\kappa\varepsilon$ along any trajectory while it remains in the tube; Proposition 5 confines every tube trajectory to $T_{C,\varepsilon}$ until it exits through $\{V = V_0 + r\}$. Hence the residence time is at most $(W_0 - \inf_T W)/(\kappa\varepsilon) \leq (\sup_T W - \inf_T W)/(\kappa\varepsilon)$. Explicitly, by L6, $\inf_T W \geq h(u_+) = L_- - c_T\varepsilon$ and $\sup_T W \leq wk/\delta + C^2/2 + c_b\sqrt{\varepsilon}$ (barrier maximal at $u = \delta$, fiber term at most $C^2/2$, cross term at most $c_b\sqrt{\varepsilon}$), so

$$\sup_T W - \inf_T W \leq \frac{wk}{\delta} - L_- + \frac{C^2}{2} + c_b\sqrt{\varepsilon} + c_T\varepsilon,$$

every term closed-form. With $\kappa = \kappa_0/2$ and $\kappa_0 = \mu_0\Gamma_{\sharp}\Gamma_b/8$, the pinned weight $w = \Gamma_{\sharp}\Gamma_b/(k\gamma_0)$ cancels $\Gamma_{\sharp}\Gamma_b$, so the leading term $(wk/\delta)/(\kappa\varepsilon)$ equals $16/(\gamma_0\mu_0\delta\varepsilon)$ exactly – the split-invariant scaling. $\square$

Remark 4b (*Division of labor*). Theorem 1's decay rate $dW/dt \leq -\kappa\varepsilon$ holds at every point of the tube, but its rated sublevel funnels form only an $O(\delta^2)$ collar hugging the cooperative face (their V -extent is $\approx C^2(x_0^2 - r)^{3/2}\gamma_0/(\Gamma_{\sharp}\Gamma_b)$, shrinking like $1/w$ at the pinned weight). Corollary 1 removes this restriction: composing the tube-wide rate with Proposition 5's confinement rates *every* tube initial condition, so the certificate covers the whole congestion tube rather than the collar alone. Every trajectory entering the strict

congestion tube exits only through the cooperative-side face $\{V = V_0 + r\}$ without fold approach. Beyond that face the trajectory enters $[V_0, V_0 + r)$, a region the full system certifies through neither result: the cooperative restoring force $\dot{V} > 0$ there is a reduced-flow fact that Section 4 declines to promote to the full 2D system. Matching the exit to the basin of Proposition 4 across $[V_0, V_0 + r)$ remains open (Remark 4a, Remark 9).

5.6 Remarks

Remark 5 (*The weight w , and the removal of the $\delta_{\min}$ threshold*). An unweighted construction fixes the slow-term weight at $w = 1$ (slow term $k/\sqrt{Q-V}$); the closure of II + IV then required $\Gamma_{\sharp}^2 < 2k\gamma_0$, producing a minimum-distance threshold $\delta_{\min}(r) = \beta k/(\alpha + \sqrt{2k\gamma_0(r)})$ and an “uncertified fold neighborhood” $[Q - \delta_{\min}^2, Q]$. That threshold was an artifact of the normalization, with a dimensional tell: the closure comparison involved k alone, while the dynamics (1)–(2) depend on (α, β, k) only through α and the product βk : two parameterizations with identical dynamics ($\beta \rightarrow \beta/c, k \rightarrow ck$) received different $\delta_{\min}$. A certificate threshold that depends on how βk is split between β and k cannot be structural.

Scaling the slow term by w removes the threshold at no cost: the cross-term cancellation, Terms I, III, V, and every ε constraint in the assembly are w -free, while the slow-closure margin $\mu_0(wk\gamma_0/4 - \Gamma_{\sharp}\Gamma_b/8)$ for the family with w free grows in w ; pinning $w = \Gamma_{\sharp}\Gamma_b/(k\gamma_0)$ globally (no $\max\{1, \cdot\}$, since the II + IV closure gives $-\Gamma_{\sharp}\Gamma_b/8$ at this value regardless and properness needs only $w > 0$) gives the w -independent $\kappa_0 = \mu_0\Gamma_{\sharp}\Gamma_b/8$ and makes the entire Lyapunov function – barrier $\Gamma_{\sharp}\Gamma_b/(\gamma_0\sqrt{Q-V})$, b , and $y^2/(2\varepsilon)$ – depend on $(\alpha, \beta k, \delta, r)$ alone, so W is itself split-invariant and the Corollary 1 bound has leading constant exactly 16 (i.e. $\lesssim 16/(\gamma_0\mu_0\delta\varepsilon)$) on the whole parameter range. The weight does double duty: the same $wk/\sqrt{Q-V}$ barrier whose δ^{-3} growth closes II + IV also pins the boundary minimum of W to the cooperative end (floor-location, §5.4), while the rated funnel’s spatial extent shrinks like $1/w$ – immaterial once Corollary 1 rates the whole tube. What survives as the limitation is the ε threshold: $\varepsilon_0(\delta, r) \rightarrow 0$ as $\delta \rightarrow 0$ (the assembly constraints scale as δ^2 and δ^4), so at fixed ε the certificate still cannot reach the fold. That joint limit, rather than any parameter-fixed distance, is the blow-up frontier of Section 6.

Remark 6 (*No parameter-fixed threshold*). For every fixed $\delta \in (0, \sqrt{x_0^2 - r})$ the certificate closes, so no parameter-dependent minimum distance from the fold survives at the level of the reduced geometry. The binding quantity is $\varepsilon_0(\delta, r)$, whose δ -scaling ($\varepsilon_0 \lesssim \delta^4$ from the residual-domination constraint $\varepsilon \leq 2\delta^4/(5\beta k)$ and the Term-II Lipschitz constraint $\varepsilon \leq (\gamma_0\delta^2/(8\beta k))^2$, both δ^4 , the latter with an r -dependent prefactor through γ_0) quantifies how the certificate degenerates as the fold is approached at fixed ε : the assembly requires $\delta \gtrsim (5\beta k\varepsilon/2)^{1/4}$. (The trade-off curve $\delta_{\min}(r)/x_0$ describes the $w = 1$ subset of the construction rather than a property of the dynamics, and is not reported as a prediction.)

Remark 7 (*Relation to Criterion 1*). Criterion 1 of Section 4.3 is the reduced-flow structural equivalence: bounded friction on the reduced flow corresponds to bounded distance from the fold. Theorem 1 and Proposition 5 are its full-system promotion under the strict-congestion truncation: bounded friction with explicit gap δ from the fold is fold-repelling for the full singularly perturbed flow, with every congestion trajectory kept at $x \geq \delta/2$ until it exits, for every $\delta > 0$, provided $\varepsilon \leq \varepsilon_0(\delta, r)$.

Remark 8 (*Why no ε -free version*). The system has two boundary regimes, the fold singularity at $x = 0$ and the cooperative equilibrium at V_0 , that each strain a different aspect of any Lyapunov certificate. A basic $W_0 = -L + y^2/(2\varepsilon)$ leaves an uncontrolled linear cross-coupling term; the linear-corrected $W_0 + b(V)y$ eliminates it; and the weighted family $w(-L) + y^2/(2\varepsilon) + b(V)y$ closes the slow-direction comparison for every δ . The resulting threshold might appear irremovable by Lyapunov engineering; it is not: the weight removes it (Remark 5). The obstruction that no choice within this family removes is the degeneration $\varepsilon_0(\delta) \rightarrow 0$ as $\delta \rightarrow 0$: every constant in the assembly is polynomial in δ , and the fast-restoration margin that powers Term I vanishes at the fold. A certificate uniform in δ at fixed ε would require resolving the singularity itself (Section 6).

Remark 9 (*Open: behavior at $V = V_0$*). Theorem 1 excludes a neighborhood $[V_0, V_0 + r)$ of the slow equilibrium. The reduced flow has a stable fixed point at (x_0, V_0) on the manifold; trajectories in the cooperative regime approach V_0 from below. A full statement combining cooperative-regime convergence to

V_0 with congestion-regime repulsion from the fold would require either matching arguments at $V = V_0$ or a global Lyapunov function valid on both regimes. This is not provided here.

Remark 10 (*Open: uniform-in- δ behavior*). The certificates apply at every fixed $\delta > 0$ but with $\varepsilon_0(\delta, r) \rightarrow 0$ as $\delta \rightarrow 0$. At fixed ε , the assembly requires $\delta \gtrsim \varepsilon^{1/4}$, so dynamics within an $O(\varepsilon^{1/4})$ -neighborhood of the fold remain uncertified. Resolving behavior there is the Krupa–Szmolyan blow-up question (Section 6), which remains open. Measured in the load variable $\mu = Q - V$, the weighted Lyapunov certificate reaches to $\mu \gtrsim \varepsilon^{1/2}$ (since $\delta \gtrsim \varepsilon^{1/4}$ and $\mu = \delta^2$ on the fold-side face), whereas for a smooth slow field the elementary reduction reaches $\mu \gtrsim \varepsilon^{2/3}$ before Krupa–Szmolyan blow-up is required. Since $1/2 < 2/3$, at $\varepsilon = 10^{-2}$ the singular field leaves $\mu \lesssim 0.10$ for the blow-up against 0.046 in the smooth case; and unlike the smooth case, where the blow-up resolves that region, here it is left open. The explicit inner-chart solution of Section 6 is the content of that $\varepsilon^{1/2}$ frontier: inside it the rescaled flow is Bernoulli-linearizable and carries orbits onto the attracting branch at Gaussian rate (Proposition 6) (in the initial-condition dependence; the slaving tail itself is algebraic, §6.2). The full chart atlas and transition maps are in place (§6.1), all three chart legs of the matching are proven (Lemmas 2–4, §6.2), and the gluing and time bookkeeping remain open (§6.2–6.3). The obstruction of Section 6 is therefore a worsening of the smooth case rather than a certificate artifact.

6. The Singularity and Its Open Resolution

The full fast-slow system (1)–(2) is singular at $x = 0$ because $g(x) = \alpha - \beta k/x$ blows up there. Standard smooth fast-slow theorems (Fenichel’s invariant manifold theorem [3] and the Krupa–Szmolyan classification of fold points [4]) require the slow vector field to be smooth in a neighbourhood of the fold; this hypothesis fails for (2). Application of the Krupa–Szmolyan apparatus to the present system therefore requires either regularising the singular slow vector field, or extending the Krupa–Szmolyan analysis to singular vector fields of the form $\alpha - \beta k/x$. Neither is performed in this paper. Multiplying the slow equation by x recasts the system in impasse form $A(x)\dot{z} = F(z)$ with impasse set $\{\det A = 0\} = \{x = 0\}$, placing it in the constrained-system programme [21], [22] and its singular-perturbation extension [23], [24]; the non-smooth alternative is the blow-up regularization of [26]. The desingularization below is that programme’s standard construction rather than a device introduced here.

The standard smooth fold blow-up is the $(1, 2, 3)$ -weighting in (x, μ, ε) with fold-centred coordinate $\mu = Q - V$ [4], whose classical inner scales are $x \sim \varepsilon^{1/3}$, $\mu \sim \varepsilon^{2/3}$; it does not accommodate the $1/x$ singularity directly. The canonical impasse desingularization [21] rescales time by $ds = dt/x$ (multiplying through by $\det A = x$, an orbital equivalence on $x > 0$), giving the polynomial system $dx/ds = x(\mu - x^2)$, $d\mu/ds = \varepsilon(\beta k - \alpha x)$, in which the $1/x$ term has resolved before blow-up. Quasi-homogeneous balance with $\deg x = 1$, $\deg \mu = 2$ makes $x\mu$ and x^3 both degree 3, forcing weight 4 on ε ; the resulting $(1, 2, 4)$ divisor field $\bar{x}' = \bar{x}\bar{\mu} - \bar{x}^3$, $\bar{\mu}' = \bar{\varepsilon}\beta k$ is regular on the whole divisor including $\bar{x} = 0$, which is what the $1/x$ term requires. Its inner region sits at $x \sim \varepsilon^{1/4}$, $\mu \sim \varepsilon^{1/2}$, matching the certificate frontier $\delta \gtrsim \varepsilon^{1/4}$, $\mu \gtrsim \varepsilon^{1/2}$ (Remark 10’s $1/2$ -vs- $2/3$ comparison is $2/a$ at $a = 4$ against the smooth $a = 3$).

The inner problem in the rescaling chart is explicitly solvable. In the chart $\bar{\varepsilon} = 1$, set $x = \varepsilon^{1/4}\bar{x}$, $\mu = \varepsilon^{1/2}\bar{\mu}$, $\sigma = \varepsilon^{1/2}s$; the chart equations are

$$\bar{x}' = \bar{x}\bar{\mu} - \bar{x}^3, \quad \bar{\mu}' = \beta k - \alpha \varepsilon^{1/4}\bar{x} = \beta k + O(\varepsilon^{1/4}),$$

recovering the divisor field above at $\bar{\varepsilon} = 1$ with the α -term entering at $O(\varepsilon^{1/4})$. The leading-order content is worth a label, with theorem language confined to the chart equation itself:

Proposition 6 (*Leading-order inner dynamics: Bernoulli linearization and ejection*). In the inner chart, at leading order in $\varepsilon^{1/4}$ (the exact chart equation with the α -correction dropped), every orbit with $\bar{x}_0 > 0$ is ejected onto the attracting branch: with $\bar{\mu}(\sigma) = \bar{\mu}_0 + \beta k \sigma$, $\bar{x}(\sigma) \rightarrow \sqrt{\bar{\mu}(\sigma)}$, universally in the initial condition, with the memory of $\bar{x}_0$ decaying at the Gaussian rate $\exp(-\beta k \sigma^2 - 2\bar{\mu}_0 \sigma)$. The statement is unconditional about the leading-order chart equation, where the Bernoulli solution is exact; it asserts nothing about the full system, whose matching across charts remains the open programme of §6.2–6.3.

Proof. At leading order $\bar{\mu}(\sigma) = \bar{\mu}_0 + \beta k \sigma$, and the $\bar{x}$ -equation is Bernoulli: the substitution $z := \bar{x}^{-2}$ linearizes it exactly,

$$z' = 2 - 2\bar{\mu}(\sigma) z,$$

a first-order linear equation with Gaussian integrating factor $\exp(2\bar{\mu}_0\sigma + \beta k \sigma^2)$, whose solution is explicit in the Dawson function (equivalently erfi). Endpoint (Watson/Laplace) asymptotics give $z(\sigma) \rightarrow 1/\bar{\mu}(\sigma)$ for every initial $z_0 > 0$, with the homogeneous part eventually decaying like $\exp(-\beta k \sigma^2 - 2\bar{\mu}_0\sigma)$ – for $\bar{\mu}_0 < 0$ a transient growth phase intervenes; the homogeneous factor falls below its initial value beyond $\sigma_0 = -2\bar{\mu}_0/(\beta k)$. Hence $\bar{x}(\sigma) \rightarrow \sqrt{\bar{\mu}(\sigma)}$: every inner orbit with $\bar{x}_0 > 0$ is ejected onto the attracting branch, universally in the initial condition and at Gaussian rate. $\square$

What Proposition 6 does and does not prove. It proves the exact transfer onto the attracting branch for the leading-order inner-chart equation only. Of the items that stood open around it, three are now closed in §6.2: the K_1 entry centre manifold (Lemma 2), the K_2 passage with uniform transient bounds and the α -correction at $O(\varepsilon^{1/4})$ (Lemma 3), and the K_3 exit landing on P_1 's centre manifold (Lemma 4). It still does not prove – and §6.2's lemmas do not supply – the gluing of the legs or the $dt = x ds$ time asymptotics on $\{x = 0\}$. Until those close, no full-system conclusion follows.

The chart problem is Bernoulli-linearizable; the smooth fold's chart- K_2 problem is Riccati, resolved through Airy functions [4]. The singular case is simpler in the inner region, not harder. The nontrivial and regular divisor dynamics confirm the $(1, 2, 4)$ weighting a posteriori: the balance chosen by quasi-homogeneity is the one at which the inner limit is neither trivial nor degenerate. The inner chart covers precisely the sliver $x \lesssim \varepsilon^{1/4}$ (load $\mu \lesssim \varepsilon^{1/2}$) that the certificate of §5 does not reach, and its dynamics place orbits on the branch (Proposition 6) that the certificate then controls. The mechanism for the uniform-in- δ statement is therefore identified, and the programme is now largely discharged: all three chart legs are theorems (Lemmas 2–4, §6.2), including the α -correction at $O(\varepsilon^{1/4})$, uniform transient bounds through the inner domain, and the landing on the exit centre manifold. What remains for the full theorem is regularity on the full exceptional divisor across the chart transitions (the gluing) and the $dt = x ds$ time bookkeeping on $\{x = 0\}$; §6.3's time accounting remains a sketch.

6.1 Blow-up chart completions and transition maps

To complete the blow-up data required for matching, we record the full desingularized vector fields in the entry chart (μ -direction) and the exit chart (x -direction), and the explicit overlap transformations between all three charts. Together with the rescaling chart already used above, this supplies the complete coordinate atlas. All fields, spectra, and transition maps below were re-derived and verified by computer algebra from the desingularized system.

Entry chart K_1 ($\bar{\mu} = 1$): coordinates $(\rho, x_1, \varepsilon_1)$ with $\mu = \rho^2$, $x = \rho x_1$, $\varepsilon = \rho^4 \varepsilon_1$; divisor $\rho = 0$. Every component of the pushed-forward field carries a factor ρ^2 ; dividing it out (desingularized time $dT = \rho^2 ds$) gives

$$\rho' = \frac{1}{2}\rho \varepsilon_1 (\beta k - \alpha \rho x_1), \quad x_1' = x_1 \left[1 - x_1^2 - \frac{1}{2}\varepsilon_1 (\beta k - \alpha \rho x_1) \right], \quad \varepsilon_1' = -2\varepsilon_1^2 (\beta k - \alpha \rho x_1).$$

On the divisor the equilibria are $E_0 = (0, 0, 0)$ with spectrum $(0, 1, 0)$ and $E_1 = (0, 1, 0)$ with spectrum $(0, -2, 0)$: both partially hyperbolic, with the centre directions along the divisor. E_1 is the divisor trace of the attracting branch on the entry side.

Exit chart K_3 ($\bar{x} = 1$): coordinates $(r_3, \mu_3, \varepsilon_3)$ with $x = r_3$, $\mu = r_3^2 \mu_3$, $\varepsilon = r_3^4 \varepsilon_3$; divisor $r_3 = 0$; desingularized time $dT = r_3^2 ds$:

$$r_3' = r_3(\mu_3 - 1), \quad \mu_3' = \varepsilon_3(\beta k - \alpha r_3) - 2\mu_3(\mu_3 - 1), \quad \varepsilon_3' = -4\varepsilon_3(\mu_3 - 1).$$

Divisor equilibria: $P_0 = (0, 0, 0)$, spectrum $(-1, 2, 4)$, fully hyperbolic; and $P_1 = (0, 1, 0)$, spectrum $(0, -2, 0)$. In K_3 the attracting branch $x = \sqrt{\mu}$ reads $\mu_3 = 1$, so P_1 is its intersection with the blow-up divisor, linearly stable in the fast direction with two centre directions.

Transition maps. With K_2 the rescaling chart ($\varepsilon = 1$; coordinates (r_2, x_2, μ_2) , and $r_2 = \varepsilon^{1/4}$ constant along orbits), on the overlap $\varepsilon_1 > 0$:

$$K_1 \rightarrow K_2 : \quad r_2 = \rho \varepsilon_1^{1/4}, \quad x_2 = x_1 \varepsilon_1^{-1/4}, \quad \mu_2 = \varepsilon_1^{-1/2},$$

and on the overlap $x_2 > 0$:

$$K_2 \rightarrow K_3 : \quad r_3 = r_2 x_2, \quad \mu_3 = \mu_2 x_2^{-2}, \quad \varepsilon_3 = x_2^{-4}.$$

All three charts share the desingularization convention $dT = r_{\text{chart}}^2 ds$, and pushing the K_1 field through the first map recovers exactly the chart field $\bar{x}' = \bar{x}\bar{\mu} - \bar{x}^3$, $\bar{\mu}' = \beta k - \alpha \varepsilon^{1/4} \bar{x}$ printed above.

6.2 The passage through the blow-up: three proven legs; the gluing open

With the atlas in hand, all three legs of the slow manifold's passage through the blow-up are now theorems: the entry leg (the centre manifold of E_1 in K_1 , Lemma 2), the passage leg (slaving, Gaussian transient, and α -correction in K_2 , Lemma 3), and the exit leg (the landing on the centre manifold of P_1 in K_3 , Lemma 4) – with each matching datum handed to the next leg in a form it absorbs exactly: Lemma 2's entry datum is the class Lemma 3's slaving remainder carries, and Lemma 3's α -term transforms to precisely the $r_3 \varepsilon_3$ coefficient of P_1 's centre manifold. All three are conventional invariant-manifold proofs – variation of constants, centre-manifold invariance equations, explicit remainder tracking at the stated orders – and the verification scripts that accompany the source machine-check every computation (chart algebra, manifold coefficients, transport laws, numerical integration of the chart fields) as a supplementary audit, not as the proofs' authority. A third tier: sixteen of the seventeen finite obligations staged from this algebra and the appendix estimands are proved in Lean 4 and kernel-checked (Appendix E) – among them Lemma 3's completed-square display, whose formal proof is then *reused* inside the formal proof of the $u = \delta$ floor, the dependency measured at the kernel proof term rather than claimed from the proof text. The one place where full Gronwall-level uniform bookkeeping is required – constants controlled simultaneously across the chart overlaps, uniformly in $\varepsilon^{1/4}$ – is exactly the open gluing step, together with the time bookkeeping of §6.3.

Lemma 2 (*K_1 entry: the centre manifold of E_1*). Let $B := \beta k$. In the entry chart K_1 , near $E_1 = (0, 1, 0)$, there exists a local two-dimensional centre manifold $W^c(E_1)$, expressible as $x_1 = 1 + h(\rho, \varepsilon_1)$ with

$$h(\rho, \varepsilon_1) = -\frac{B}{4} \varepsilon_1 + \frac{\alpha}{4} \rho \varepsilon_1 - \frac{9B^2}{32} \varepsilon_1^2 + O_3,$$

where O_3 collects terms of total order at least three in (ρ, ε_1) ; in particular $h = -\frac{B}{4} \varepsilon_1 + O(\rho \varepsilon_1, \varepsilon_1^2)$, and $h(\rho, 0) \equiv 0$ (the manifold contains the critical branch). Under the overlap map from K_1 to K_2 ,

$$x_2 = x_1 \varepsilon_1^{-1/4}, \quad \mu_2 = \varepsilon_1^{-1/2},$$

the transported branch satisfies

$$x_2^2 = \mu_2 - \frac{B}{2\mu_2} + O\left(\frac{\rho}{\mu_2} + \mu_2^{-3}\right),$$

so $x_2^2/\mu_2 \rightarrow 1$ on the K_1 – K_2 overlap as $\mu_2 \rightarrow +\infty$. On a finite matching section $\mu_2 = M$,

$$x_2^2 = M - \frac{B}{2M} + O\left(\frac{\rho}{M} + M^{-3}\right).$$

On the divisor ($\rho = 0$), or along any overlap family with $\rho = O(M^{-2})$, this reduces to $x_2^2 = M - \frac{B}{2M} + O(M^{-3})$. Along the natural entry family, where the section sits at $\rho = r M^{1/2}$ (from $r = \rho \varepsilon_1^{1/4}$ at $\varepsilon_1 = M^{-2}$), the matching datum is

$$x_2^2 = M - \frac{B}{2M} + O(M^{-3} + r M^{-1/2}), \quad \text{equivalently} \quad w(0) = \frac{B}{2M^2} + O(M^{-4} + r M^{-3/2}),$$

whose r -term lies in the same class as the K_2 passage lemma's $O(r \mu_2^{-3/2})$ slaving remainder and is absorbed by it.

Proof. Set $u = x_1 - 1$. Blowing up the desingularized field $x' = x(\mu - x^2)$, $\mu' = \varepsilon(B - \alpha x)$, $\varepsilon' = 0$ by $x = \rho x_1$, $\mu = \rho^2$, $\varepsilon = \rho^4 \varepsilon_1$, and dividing by ρ^2 , the K_1 equations are

$$\begin{aligned}\rho' &= \tfrac{1}{2} \rho \varepsilon_1 (B - \alpha \rho(1 + u)), & u' &= (1 + u) \left[-2u - u^2 - \tfrac{1}{2} \varepsilon_1 (B - \alpha \rho(1 + u)) \right], \\ \varepsilon_1' &= -2 \varepsilon_1^2 (B - \alpha \rho(1 + u)).\end{aligned}$$

Linearizing at $(\rho, u, \varepsilon_1) = (0, 0, 0)$ gives

$$J = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -2 & -\frac{1}{2}B \\ 0 & 0 & 0 \end{pmatrix},$$

with spectrum $\{0, -2, 0\}$ and centre eigenspace $E^c = \{u = -\frac{B}{4} \varepsilon_1\}$. By the Centre Manifold Theorem there is a local invariant two-dimensional manifold tangent to E^c ; take $W^c(E_1)$ to be its branch-continuation representative, $u = h(\rho, \varepsilon_1)$, so $h(0, 0) = 0$, $h_\rho(0, 0) = 0$, $h_{\varepsilon_1}(0, 0) = -\frac{B}{4}$. The plane $\{\varepsilon_1 = 0, u = 0\}$ is invariant, so $h(\rho, 0) \equiv 0$ and h carries no pure- ρ terms. Substituting the ansatz $h = a_1 \varepsilon_1 + b_1 \rho \varepsilon_1 + c_1 \varepsilon_1^2 + O_3$ into the invariance equation

$$h_\rho \rho' + h_{\varepsilon_1} \varepsilon_1' = (1 + h) \left[-2h - h^2 - \tfrac{1}{2} \varepsilon_1 (B - \alpha \rho(1 + h)) \right]$$

and matching coefficients gives

$$a_1 = -\frac{B}{4}, \quad b_1 = \frac{\alpha}{4}, \quad c_1 = -\frac{9B^2}{32},$$

with the residual of total order three. Hence $x_1 = 1 - \frac{B}{4} \varepsilon_1 + O(\rho \varepsilon_1, \varepsilon_1^2)$, so $x_1^2 = 1 - \frac{B}{2} \varepsilon_1 + \frac{\alpha}{2} \rho \varepsilon_1 - \frac{B^2}{2} \varepsilon_1^2 + O_3$. Applying the transition map, $x_2^2 = x_1^2 \varepsilon_1^{-1/2}$ with $\varepsilon_1 = \mu_2^{-2}$:

$$x_2^2 = \mu_2 \left[1 - \frac{B}{2} \mu_2^{-2} + \frac{\alpha}{2} \rho \mu_2^{-2} + O(\mu_2^{-4}) \right] = \mu_2 - \frac{B}{2\mu_2} + O\left(\frac{\rho}{\mu_2} + \mu_2^{-3}\right).$$

The section statements follow by evaluating at $\mu_2 = M$; on the natural family $\rho = rM^{1/2}$, the ρ -term is $\frac{\alpha}{2} r M^{-1/2}$. $\square$

Lemma 3 (*K_2 passage: Gaussian transient, slaving tail, and α -correction*). Let $B := \beta k$ and $r := r_2 = \varepsilon^{1/4}$. In the rescaling chart K_2 the desingularized equations are

$$x_2' = x_2 (\mu_2 - x_2^2), \quad \mu_2' = B - \alpha r x_2, \quad r' = 0.$$

Set

$$z := x_2^{-2}, \quad w := \mu_2 z - 1,$$

so that w satisfies the exact equation

$$w' = -2\mu_2 w + (B - \alpha r x_2) z, \quad z = \frac{1 + w}{\mu_2}.$$

Assume Lemma 2 supplies the matching data at the section $\mu_2(0) = M$:

$$x_2^2(0) = M - \frac{B}{2M} + O(M^{-3} + r M^{-1/2}), \quad \text{equivalently} \quad w(0) = \frac{B}{2M^2} + O(M^{-4} + r M^{-3/2})$$

(the r -term is the entry family's α -trace, in the same class as the slaving remainder below and absorbed by it; on the divisor it vanishes). Write $A(\sigma) := M + B\sigma$, and let $S = S(r)$ be a passage time with

$$A_S := A(S) \rightarrow +\infty, \quad r A_S^{1/2} \rightarrow 0 \quad (r \rightarrow 0);$$

the inner-domain scale $S = o(r^{-1}) = o(\varepsilon^{-1/4})$ satisfies this. Then, uniformly for $0 \leq \sigma \leq S$:

1. (*Gaussian decay of entry memory.*) The homogeneous dependence on the entry datum obeys

$$|T(\sigma)| \leq C \left| w(0) - \frac{B}{2M^2} \right| \exp(-2M\sigma - B(1 - o(1))\sigma^2),$$

and for $r = 0$ the homogeneous solution is exactly $\exp(-2M\sigma - B\sigma^2) (1 + B\sigma/M)$: the memory of the K_1 entry decays at the Gaussian rate $\exp(-B\sigma^2 - 2M\sigma)$.

2. (*Algebraic slaving tail.*)

$$w(\sigma) = \frac{B}{2\mu_2(\sigma)^2} + O(\mu_2(\sigma)^{-3} + r\mu_2(\sigma)^{-3/2}) + T(\sigma),$$

equivalently

$$x_2^2(\sigma) = \mu_2(\sigma) - \frac{B}{2\mu_2(\sigma)} + O(\mu_2(\sigma)^{-2} + r\mu_2(\sigma)^{-1/2}) + O(\mu_2(\sigma)T(\sigma));$$

in particular $x_2^2/\mu_2 \rightarrow 1$ uniformly on the matching region $\mu_2 \rightarrow +\infty$, $r\sqrt{\mu_2} \rightarrow 0$.

3. (*Uniform α -correction.*)

$$\mu_2(\sigma) = A(\sigma) + r\mu_2^{(1)}(\sigma) + R_\mu(\sigma), \quad \mu_2^{(1)}(\sigma) = -\frac{2\alpha}{3B} \left[(M + B\sigma)^{3/2} - M^{3/2} \right],$$

with

$$\sup_{0 \leq \sigma \leq S} |R_\mu(\sigma)| \leq C(rM^{-1/2} + r^2A_S^2), \quad \text{so} \quad \frac{\mu_2(\sigma)}{A(\sigma)} = 1 + O(rA(\sigma)^{1/2}) + o(1)$$

uniformly on $0 \leq \sigma \leq S$: on every passage scale $S = o(r^{-1})$ the α -correction stays uniformly controlled through the whole inner domain.

At an exit section $\sigma = S$ with $\mu_2(S) \rightarrow +\infty$ and $r\sqrt{\mu_2(S)} \rightarrow 0$, the $K_2 \rightarrow K_3$ transition gives

$$r_3 = r x_2 \rightarrow 0, \quad \mu_3 = \mu_2 x_2^{-2} = 1 + w \rightarrow 1, \quad \varepsilon_3 = x_2^{-4} \rightarrow 0 :$$

the orbit exits K_2 into K_3 at the divisor trace $P_1 = (0, 1, 0)$ of the attracting branch.

Proof. The K_2 equations give $z' = -2\mu_2 z + 2$; since $w = \mu_2 z - 1$,

$$w' = \mu_2' z + \mu_2 z' = (B - \alpha r x_2) z + \mu_2(-2\mu_2 z + 2) = -2\mu_2 w + (B - \alpha r x_2) z,$$

the exact w -equation. From the entry data, $x_2^2(0) = M(1 - B/(2M^2) + O(M^{-4}))$, so $z(0) = M^{-1}(1 + B/(2M^2) + O(M^{-4}))$ and $w(0) = Mz(0) - 1 = B/(2M^2) + O(M^{-4})$.

Gaussian transient. Variation of constants gives

$$w(\sigma) = \Phi(\sigma, 0) w(0) + \int_0^\sigma \Phi(\sigma, \tau) (B - \alpha r x_2(\tau)) z(\tau) d\tau, \quad \Phi(\sigma, \tau) = \exp\left(-2 \int_\tau^\sigma \mu_2(s) ds\right).$$

On the inner passage $r\sqrt{\mu_2} \rightarrow 0$, so $\mu_2' = B(1 + o(1))$ uniformly, hence $\mu_2(\sigma) = M + B\sigma + o(B\sigma)$ and $\int_0^\sigma \mu_2 = M\sigma + \frac{B}{2}\sigma^2 + o(\sigma^2)$, giving $\Phi(\sigma, 0) = \exp(-2M\sigma - B(1 - o(1))\sigma^2)$. For $r = 0$, $\mu_2 = M + B\sigma$ exactly and $\Phi(\sigma, 0) = \exp(-2M\sigma - B\sigma^2)$; the homogeneous equation for a difference of solutions, $\Delta w' = (-2\mu_2 + B/\mu_2)\Delta w$, integrates in closed form to $\Delta w(\sigma) = \Delta w(0) \exp(-2M\sigma - B\sigma^2) (1 + B\sigma/M)$, which the stated bound absorbs.

Algebraic slaving. With $z = (1 + w)/\mu_2$ the w -equation reads $w' = -2\mu_2 w + \mu_2^{-1}(B - \alpha r x_2)(1 + w)$. The stable coefficient $-2\mu_2$ forces w onto the instantaneous slaving scale; the leading balance $0 = -2\mu_2 w + B/\mu_2$ gives $w_{\text{sl}} = B/(2\mu_2^2)$. The $\alpha r x_2$ part contributes $O(r\mu_2^{-3/2})$ (since $x_2 \sim \sqrt{\mu_2}$), and differentiating the slaving ansatz contributes the next algebraic correction $(-w'_{\text{sl}}/(2\mu_2) = B^2/(2\mu_2^4))$, within the stated $O(\mu_2^{-3})$. Variation of constants applied to $w - B/(2\mu_2^2)$ then yields estimate 2, with T bounded by the Gaussian transient.

Since $x_2^2 = \mu_2/(1+w) = \mu_2(1-w+O(w^2))$, substituting the slaving estimate gives the x_2^2 expansion, and $x_2^2/\mu_2 \rightarrow 1$ uniformly in the overlap.

α -correction. From the slaving estimate, $x_2 = \sqrt{\mu_2} (1 + O(\mu_2^{-2} + r\mu_2^{-3/2}) + O(T))$, so

$$\mu_2' = B - \alpha r \sqrt{\mu_2} + O(r\mu_2^{-3/2}) + O(r^2\mu_2^{-1}) + O(r\sqrt{\mu_2}T).$$

Writing $\mu_2 = A(\sigma) + r\nu(\sigma)$, the first order in r is $\nu' = -\alpha\sqrt{A(\sigma)}$, and with $\nu(0) = 0$,

$$\nu(\sigma) = -\alpha \int_0^\sigma \sqrt{M+B\tau} d\tau = -\frac{2\alpha}{3B} \left[(M+B\sigma)^{3/2} - M^{3/2} \right] = \mu_2^{(1)}(\sigma).$$

The residual terms integrate uniformly to $R_\mu = O(rM^{-1/2}) + O(r^2A_S^2)$ on $0 \leq \sigma \leq S$, and since $rA_S^{1/2} \rightarrow 0$ the relative correction is $r\mu_2^{(1)}/A = O(rA^{1/2}) = o(1)$ uniformly.

Exit. The $K_2 \rightarrow K_3$ transition is $r_3 = rx_2$, $\mu_3 = \mu_2x_2^{-2}$, $\varepsilon_3 = x_2^{-4}$. Using $x_2^2 \sim \mu_2$: $r_3 = r\sqrt{\mu_2}(1+o(1)) \rightarrow 0$, $\mu_3 = \mu_2z = 1+w \rightarrow 1$, and $\varepsilon_3 \sim \mu_2^{-2} \rightarrow 0$. The transported orbit exits K_2 into K_3 at $P_1 = (0, 1, 0)$. $\square$

Two smallness mechanisms, not one. Lemma 3 separates them precisely: the memory of the entry datum is Gaussian-small ($\exp(-B\sigma^2 - 2M\sigma)$, and exactly $\exp(-2M\sigma - B\sigma^2)(1 + B\sigma/M)$ at $r = 0$), while the persistent forcing leaves the *algebraic* slaving tail $w = B/(2\mu_2^2) + O(\mu_2^{-3} + r\mu_2^{-3/2})$. Claims of exponentially small deviation from the branch would be false; only the entry memory is Gaussian-small.

Lemma 4 (K_3 exit: landing on the centre manifold of P_1). Let $B := \beta k$. In the exit chart K_3 ($x = r_3$, $\mu = r_3^2\mu_3$, $\varepsilon = r_3^4\varepsilon_3$), the desingularized equations are

$$r_3' = r_3(\mu_3 - 1), \quad \mu_3' = \varepsilon_3(B - \alpha r_3) - 2\mu_3(\mu_3 - 1), \quad \varepsilon_3' = -4\varepsilon_3(\mu_3 - 1),$$

and $r_3^4\varepsilon_3 = \varepsilon$ is conserved exactly. The point $P_1 = (r_3, \mu_3, \varepsilon_3) = (0, 1, 0)$ is an equilibrium with spectrum $\{0, -2, 0\}$; by the Centre Manifold Theorem it carries a local two-dimensional centre manifold $W^c(P_1)$, expressible as $\mu_3 = 1 + h_3(r_3, \varepsilon_3)$ with

$$h_3(r_3, \varepsilon_3) = \frac{B}{2}\varepsilon_3 - \frac{\alpha}{2}r_3\varepsilon_3 + \frac{B^2}{4}\varepsilon_3^2 + O_3, \quad h_3(r_3, 0) \equiv 0$$

(the manifold contains the branch axis $\{\varepsilon_3 = 0, \mu_3 = 1\}$). Un-blowing, $W^c(P_1)$ is exactly the classical first-order slow manifold of the desingularized system:

$$\mu = x^2 + \varepsilon \frac{B - \alpha x}{2x^2} + O(\varepsilon^2).$$

The orbit transported by Lemma 3 arrives, at an exit section with $\mu_2 \rightarrow \infty$ and $r\sqrt{\mu_2} \rightarrow 0$, with

$$\mu_3 - 1 - h_3(r_3, \varepsilon_3) = O(\varepsilon_3^{3/2}) + T,$$

where the coefficients of ε_3 and $r_3\varepsilon_3$ in the arrival data match h_3 exactly – Lemma 3's slaving α -term transforms to precisely the manifold's $-\frac{\alpha}{2}r_3\varepsilon_3$ – so the stated bound is inherited from Lemma 3's algebraic remainder, and T is the Gaussian-small entry memory. The transverse eigenvalue -2 contracts this mismatch exponentially in the desingularized time while, on $W^c(P_1)$, the flow

$$\varepsilon_3' = -2B\varepsilon_3^2 + O(\varepsilon_3^3 + r_3\varepsilon_3^2), \quad \frac{r_3'}{r_3} = \frac{B}{2}\varepsilon_3 + O(r_3\varepsilon_3 + \varepsilon_3^2)$$

carries the orbit out of the blow-up region: ε_3 decays algebraically, $r_3 = x$ grows, and $r_3^4\varepsilon_3 = \varepsilon$ holds exactly throughout. At any fixed exit section $x = \rho_{\text{out}}$ the orbit is $O(\varepsilon^{3/2})$ -close (constant depending on ρ_{out}) to the slow manifold $\mu = x^2 + \varepsilon(B - \alpha x)/(2x^2)$: the hand-off to the outer (Fenichel) region on the attracting branch.

Proof. Blowing up the desingularized field $x' = x(\mu - x^2)$, $\mu' = \varepsilon(B - \alpha x)$, $\varepsilon' = 0$ by $x = r_3$, $\mu = r_3^2\mu_3$, $\varepsilon = r_3^4\varepsilon_3$ and dividing by r_3^2 gives the displayed equations; $\frac{d}{ds}(r_3^4\varepsilon_3) = 4r_3^4\varepsilon_3(\mu_3 - 1) - 4r_3^4\varepsilon_3(\mu_3 - 1) = 0$ exactly.

Linearizing at P_1 gives $J = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -2 & B \\ 0 & 0 & 0 \end{pmatrix}$ with centre eigenspace $\mu_3 - 1 = \frac{B}{2}\varepsilon_3$ (r_3 free). The plane $\{\varepsilon_3 = 0, \mu_3 = 1\}$ is invariant, so the branch-continuation representative of $W^c(P_1)$ has $h_3(r_3, 0) \equiv 0$ and no pure- r_3 terms. Substituting $h_3 = a_3\varepsilon_3 + b_3r_3\varepsilon_3 + c_3\varepsilon_3^2 + O_3$ into the invariance equation $h_{3,r_3}r'_3 + h_{3,\varepsilon_3}\varepsilon'_3 = \mu'_3|_{\mu_3=1+h_3}$ and matching coefficients gives $a_3 = B/2$, $b_3 = -\alpha/2$, $c_3 = B^2/4$, with residual of total order three. Un-blowing with $r_3 = x$, $\varepsilon_3 = \varepsilon/x^4$, $\mu_3 = \mu/x^2$:

$$\mu = x^2(1 + h_3) = x^2 + \varepsilon \frac{B - \alpha x}{2x^2} + \frac{B^2}{4} \frac{\varepsilon^2}{x^6} + O_3,$$

which is the classical first-order slow manifold (obtained independently from the invariance relation $x(\mu - x^2) = \frac{dx}{d\mu} \varepsilon(B - \alpha x)$ with $\mu = x^2 + \varepsilon\varphi(x)$: $\varphi = (B - \alpha x)/(2x^2)$).

For the landing, Lemma 3 gives the arrival data $w = \frac{B}{2}\mu_2^{-2} - \frac{\alpha}{2}r\mu_2^{-3/2} + O(\mu_2^{-3}) + T$, and the $K_2 \rightarrow K_3$ transition gives $\varepsilon_3 = \mu_2^{-2}(1 + O(w))$, $r_3 = r\sqrt{\mu_2}(1 + O(w))$, $\mu_3 - 1 = w$. Substituting,

$$h_3(r_3, \varepsilon_3) = \frac{B}{2}\mu_2^{-2} - \frac{\alpha}{2}r\mu_2^{-3/2} + O(\mu_2^{-4} + r\mu_2^{-5/2}),$$

so $\mu_3 - 1 - h_3 = O(\mu_2^{-3}) + T = O(\varepsilon_3^{3/2}) + T$: both tracked orders cancel identically and only Lemma 3's remainder survives. Near P_1 the transverse direction contracts at rate $-2 + o(1)$, so the mismatch decays exponentially in s while the centre flow, obtained by substituting h_3 into the r_3 and ε_3 equations, is as displayed. Along it ε_3 is strictly decreasing (for $\varepsilon_3 > 0$, $\mu_3 > 1$ on the manifold near P_1), r_3 is strictly increasing, and the conserved $r_3^4\varepsilon_3 = \varepsilon$ converts the exit section $r_3 = \rho_{\text{out}}$ into $\varepsilon_3 = \varepsilon/\rho_{\text{out}}^4$; un-blowing the contracted mismatch bound there gives the stated $O(\varepsilon^{3/2})$ closeness to $\mu = x^2 + \varepsilon(B - \alpha x)/(2x^2)$. $\square$

What remains: the gluing. Lemmas 2–4 prove the three legs separately, each with explicit constants and matched hand-off data. The remaining step is the gluing: a single statement, uniform in $\varepsilon^{1/4}$, that the transported manifold exists, is smooth, and satisfies the three estimates simultaneously with constants controlled across the chart overlaps. Nothing in the gluing is a theorem here; it is the sole remaining carrier of the uniform-in- δ statement (together with the §6.3 time bookkeeping, which stays a sketch).

6.3 Time bookkeeping (formal sketch)

The original time satisfies $dt = x ds$, which degenerates on $\{x = 0\}$. Per chart, with the shared convention $dT = r_{\text{chart}}^2 ds$: in K_1 , $dt = (x_1/\rho) dT$, so the original time diverges only if the orbit is followed to the divisor itself (backwards); in K_2 , $dt = (x_2/r_2) d\sigma$, so along the slaved family

$$t_{K_2} \sim \varepsilon^{-1/4} \frac{2}{3\beta k} \left[(M + \beta k \sigma_{\text{exit}})^{3/2} - M^{3/2} \right],$$

which is the dominant contribution, $O(\varepsilon^{-1/4})$ when the exit boundary is ε -independent; in K_3 , $dt = dT/r_3$ with r_3 bounded away from zero at exit, so the post-blow-up time is regular. For every fixed $\varepsilon > 0$ the total original time along an orbit with $x > 0$ is finite. All scaling statements here are sketches: they presuppose the matching of §6.2, and the $\varepsilon \rightarrow 0$ asymptotics of t require a precise choice of entry/exit boundaries and a rigorous matched expansion, none of which is performed here.

With the chart fields and transition maps of §6.1 in hand, and all three chart legs now proven (Lemmas 2–4, §6.2, including the centre-manifold arguments at both partially hyperbolic divisor equilibria E_1 and P_1), a Krupa–Szmolyan jump-point classification still requires the gluing of the legs and the divisor-to-original-time bookkeeping beyond the scaling sketch of §6.3 (the rescaling degenerates on $\{x = 0\}$). None of this is performed here as a theorem; the Section 1 Status of Claims table records the blow-up question as open. Closing this item is the natural follow-up to the present paper and would, if successful, resolve the uniform-in- δ behavior at fixed ε left open in Section 5 (Remark 10).

This programme is active. The equivalence between impasse dynamics and multiple-timescale singular perturbation is developed in [23], [24], and a recent planar parasite–host model [25] resolves a frequency-dependent extinction singularity by the blow-up method, with a regime split ($\beta - d$, equivalently $\mathcal{R}_0 - 1$, of

order 1 versus order ε) parallel to the cooperative/congestion partition here. Those analyses are qualitative – orbit passage, homoclinic families, and the canard-like or *funneling* trapping of orbits near the repelling manifold [25] – in ecological and epidemiological settings, with order-of-magnitude ($O(1)$, $O(\varepsilon)$) scales and, at most, a LaSalle argument rather than a finite- ε estimate. The present paper contributes the quantitative finite- ε certificate in the capacity-constrained setting: the funnel here is a sublevel set of an explicit Lyapunov function with a written-down decay rate and exit-time bound, a distinct object from the qualitative canard funneling of [25], and the pole-field blow-up itself is left open.

7. Empirical Predictions and Identifiability

Scope and relation to the certificates. Theorem 1 and Proposition 5 certify fold repulsion and finite-time exit on the strict congestion subsets $J(r, \delta) = [V_0 + r, Q - \delta^2]$ for *every* fixed $\delta > 0$, with the smallness threshold $\varepsilon_0(\delta, r)$ degenerating as $\delta \rightarrow 0$ (Remark 10). The predictions below concern $V \rightarrow Q^-$, the pre-transition regime under *exogenous* forcing. Proposition 4 supplements the certificates with local stability of the cooperative equilibrium (x_0, V_0) ; the matching between Proposition 4’s basin and the certified tube is open (Remark 4a). Predictions P1–P4 do not depend on Theorem 1, Proposition 4, or Proposition 5; prediction P5 below tests the barrier certificate directly.

Prediction hierarchy. The tests are ordered by how directly they bear on the model’s distinctive content. *Primary:* P4 tests the fold classification itself — the derived saddle-node scaling with OU noise — and Appendix A gives its protocol; the friction-exponent measurement of Proposition 2 tests the hyperbolic coupling $F = k/x$ that places the model outside the smooth theory; P5 tests the autonomous barrier mechanism, the certificate’s empirical shadow. *Secondary:* P1–P2 are generic fold signatures shared with any saddle-node mechanism, and P3 is conjectural (its recovery path is uncertified); they discriminate the framework from null models but not from competing fold mechanisms.

The division of labor is structural. The certificates concern the *autonomous* slow dynamics: the friction barrier of Proposition 3 lifts to the full singular flow, so endogenous dynamics near the slow manifold cannot reach the fold from any fixed distance (at sufficiently small ε); off-manifold transients lie outside the certificate. The empirical predictions P1–P4 are about *exogenous* fold approach: a process outside the autonomous slow equation (external load growth, structural capacity reduction, quasi-static perturbation) drives V toward Q . The CSD signatures are derived from the saddle-node linearisation of the fast subsystem around the attracting branch $x^*(V) = \sqrt{Q - V}$, which exists for every $V < Q$; the linearisation requires none of the certificates and applies on the whole branch.

The two results combine to tell the same story from two sides: autonomous dynamics near the slow manifold cannot reach the fold within the certified tube (the friction barrier holds), and when something else forces the system toward Q , the pre-transition signature is fold-consistent CSD (P1–P4). The blow-up of Section 6 would extend the certified region uniformly in δ ; pending that, the present split is the conservative accounting of what each result establishes.

We state four fold-consistent predictions for exogenous approach (P1–P4), with explicit identifiability and competing mechanisms, and one barrier prediction for the autonomous regime (P5).

P1 (thresholded transition). The regime transition is thresholded rather than gradual. *Falsified if:* observed transitions are smooth and proportional to load. *Competing mechanisms to rule out:* gradual capacity degradation, smooth performance decline.

P2 (fold signature). Pre-transition dynamics are dominated by real-eigenvalue (saddle-node) signatures rather than complex-eigenvalue (Hopf) signatures. *Falsified if:* observed pre-transition variability shows cyclical instability with complex eigenvalue dominance. *Competing mechanisms to rule out:* oscillatory bottlenecks, periodic forcing, delay-induced oscillation.

P4 (fold-consistent scaling). Near transition, recovery time and stationary fluctuation variance scale with the same fold exponent:

$$\tau \propto |\mu|^{-1/2}, \quad \text{Var} \propto |\mu|^{-1/2},$$

where $\mu = Q - V$. *Falsified if:* observed exponents are systematically different from $-1/2$ for both observables, or if τ and Var exhibit systematically different exponents. *Competing mechanisms to rule out:* nonnormality, delay-induced critical slowing, Hopf proximity (giving $\tau \propto |\mu|^{-1}$), noise-driven near-bifurcation behaviour without underlying saddle-node.

Derivation of P4. Linearising the fast subsystem around the attracting branch, $\delta\dot{x} = f_x(x^*, V) \delta x = -2x^* \delta x = -2\sqrt{\mu} \delta x$. The relaxation eigenvalue is $\lambda(\mu) = 2\sqrt{\mu}$. Standard Ornstein–Uhlenbeck analysis under additive noise of constant intensity D gives:

$$\tau(\mu) = \frac{1}{\lambda(\mu)} = \frac{1}{2\sqrt{\mu}}, \quad \text{Var}(\delta x) = \frac{D}{2\lambda(\mu)} = \frac{D}{4\sqrt{\mu}}.$$

Both observables share the exponent $-1/2$. This exponent is the universal saddle-node CSD signature; it does not depend on the friction-coupling form $F = k/x^a$. The friction-exponent prediction of Proposition 2 (which does depend on a) is a distinct claim requiring direct measurement of per-unit coordination cost and is not tested by the protocol of Appendix A. The ratio $\text{Var}/\tau = D/2$ is independent of μ and provides an internal cross-check: under the fold mechanism with constant-intensity noise, the two CSD indicators must scale with the same exponent. Departures from this proportionality indicate either non-constant noise intensity or a different bifurcation class.

The empirical protocol for testing P4 is specified in Appendix A.

Remark 11. We use *fold-consistent* prediction rather than *fold* prediction to mark that the observed patterns are consistent with the fold model but do not uniquely identify it without ruling out the listed competitors.

P5 (barrier: autonomous relaxation). Predictions P1–P4 concern exogenous fold approach and do not test Theorem 1. The friction-barrier certificate makes a separate, testable prediction. In the strict congestion regime $V_0 < V < Q$ the autonomous slow flow points *away* from the fold, toward the standoff V_0 ; so a capacity-constrained system in that regime, with exogenous drivers held stationary, relaxes toward V_0 rather than drifting to Q . *Falsified if:* the slow variable is observed in autonomous monotone approach $V \rightarrow Q$ through the congestion regime while the exogenous controls (α, β, k, Q) are demonstrably stationary across the window; the falsifying observation must lie in the certified region $[V_0 + r, Q - \delta^2]$ with the state near its slow manifold, excluding the uncertified sliver $\mu \lesssim \varepsilon^{1/2}$ of Remark 10 and far-off-manifold transients.

Operationalisation (the identification bottleneck). The barrier is compatible with fold approach *under forcing*; only autonomous approach falsifies it, so a valid test must certify driver-stationarity. This is the same exogenous-drive-versus-autonomous-drift distinction that §A.6 already carries for P1–P4. Two designs are admissible: (a) a naturally driver-stationary window, in which the controls are measured or argued constant and the sign of $\dot{V}$ in the congestion regime is read directly; (b) a matched-control comparison of $\dot{V}$ under a known-forcing condition against a no-forcing condition, the barrier predicting relaxation only in the latter. Absent driver-stationarity the observation is uninformative about the barrier. P5 is therefore falsifiable in principle and demanding in practice: it states the observable that makes the certificate earn its empirical keep, and defers closing the operationalisation to the identification programme of §A.6.

Possible extension: the damage-memory prediction. P3 is kept apart from P1, P2, P4, and P5 because it conjectures dynamics the autonomous model does not certify: the model proves a stable standoff, not a hysteresis loop, and delayed recovery is a hysteresis-type signature. Its promotion path is stated in its own terms.

P3 (delayed recovery, conjectural). The transition is abrupt and recovery is delayed: returning to admissible operation requires V significantly below the collapse threshold. *Falsified if:* recovery occurs at the same load level as collapse with no delay or overshoot. *Competing mechanisms to rule out:* hysteresis from independent causes, memory effects in load measurement. *Promotion conditions:* P3 would move

from conjectural to derived if a slow-equation augmentation modelling structural damage (state-dependent decrement of Q following a fold crossing) were specified and shown to produce a hysteresis loop with a quantitatively predicted minimum separation between collapse and recovery loads.

8. Discussion

8.1 What the Model Contributes

1. **An explicit finite- ε certificate (principal contribution).** Theorem 1 and Proposition 5 give a quantitative Lyapunov/barrier funnel with every constant written down: on strict congestion subsets the flow is fold-repelling with reach $\delta \gtrsim \varepsilon^{1/4}$ (load $\mu \gtrsim \varepsilon^{1/2}$), net decay rate $\kappa = \kappa_0/2$, and a closed-form exit-time bound $\lesssim 1/(\kappa\varepsilon)$. The constrained/impassé and singular-perturbation literature treats this class of singularities by qualitative blow-up (phase portraits, orbit passage, canards); a finite- ε certificate with explicit reach and exit time is the methodological difference here.
2. **A fold on an impassé set.** Multiplying the slow equation by x puts the system in impassé form $A(x)\dot{z} = F(z)$ with impassé set $\{\det A = 0\} = \{x = 0\}$; the configuration situates within the constrained-system and impassé-plus-singular-perturbation programme [21]–[24], and the $ds = dt/x$ desingularization of §6 is that programme’s canonical move rather than a new device. What is specific here is the pole field $g(x) = \alpha - \beta k/x$: its desingularized branch flow $\dot{x} = (\beta k - \alpha x)/(2x^2)$ carries a double pole with fixed repelling sign, a different local object from the impassé normal forms classified in [22].
3. **The reach exponent as a sharpness signal.** The load reach $\mu \gtrsim \varepsilon^{1/2}$, against $\varepsilon^{2/3}$ for a smooth slow field, coincides after §6 with the $(1, 2, 4)$ -versus- $(1, 2, 3)$ quasi-homogeneous inner scales – evidence that the reach is set by the pole rather than by the Lyapunov family. The explicit inner solution of §6 upgrades the weighting coincidence from evidence to demonstration in the rescaling chart, and the chart atlas of §6.1 extends the demonstration to the entry and exit charts.
4. **A self-organized operating margin and dynamic fold-shielding.** The stable standoff from the critical point $x = 0$ is $x_0 = \beta k/\alpha$ (with $V_0 = Q - x_0^2$), the growth-to-friction ratio. Within the certified tube (fixed δ , $\varepsilon \leq \varepsilon_0$ and $\varepsilon < \varepsilon_{\text{blk}}$) the autonomous flow is fold-repelling, so the fold is reachable only under exogenous forcing (P5). The claim is conditional: off-tube transients, the $\mu \lesssim \varepsilon^{1/2}$ sliver, and $[V_0, V_0 + r)$ are uncertified, and with the drivers (Q, α, β, k) frozen the Ashwin [16] B-tipping notion, which needs a drifting parameter, does not literally apply.
5. **The certified-versus-open boundary.** The certificate of item 1 rests on the admissibility equivalence (barrier monotonicity, Proposition 3; Criterion 1) and an isolating-block certificate covering all tube initial conditions; its threshold $\varepsilon_0(\delta, r)$, with δ^4 scaling, locates the certified-versus-open boundary at the joint limit $\delta \rightarrow 0$, ε fixed.
6. **A conditional scaling law.** The friction-divergence rate $F \sim (Q - V)^{-1/2}$ on the attracting branch is a derived, conditional, falsifiable prediction (Proposition 2); the underlying $a = 1$ coupling has a queueing pedigree [20].

Convergent rediscovery. The same fold-with-CSD structure has been identified independently in ecological tipping-point research (slow-recovery indicators as generic precursors of nearby catastrophic shifts [7], rising variance as a leading indicator of ecological transition [9]) and in climate-bifurcation detection (degenerate fingerprinting applied to the Atlantic Meridional Overturning Circulation [8]), within the critical-transitions framework [11] whose critical-slowness-down lineage traces to [19]. The friction-barrier mechanism and the Lyapunov certificate developed here are framework contributions specific to the capacity-constrained setting; the CSD scaling laws they imply belong to a longer convergent tradition.

8.2 Open Issues and Scope

1. The full fast-slow system is singular and the Krupa–Szmolyan classification is not applied as a theorem; the certificates of Section 5 hold at every fixed $\delta > 0$ but with $\varepsilon_0(\delta, r) \rightarrow 0$ as $\delta \rightarrow 0$, so uniform-in- δ behavior at fixed ε remains open.
2. Proposition 6 is a leading-order inner-chart statement, now flanked by the proven K_1 entry, K_2 passage, and K_3 exit-landing legs (Lemmas 2–4, §6.2, with uniform transient bounds, the α -correction at $O(\varepsilon^{1/4})$, and the un-blown identification of the exit manifold with the classical slow manifold); the gluing of the

legs and the $dt = x ds$ time asymptotics remain open, and no full-system conclusion follows until they close.

3. The Krupa–Szmolyan classification is not performed as a theorem; Section 6 solves the rescaling-chart inner problem, derives the full chart atlas and transition maps (§6.1), and sketches the matching, α -correction, and time bookkeeping (§6.2–6.3); the remaining items are the rigorous closure of those sketches: centre-manifold existence through the blow-up, uniform transient bounds, and time asymptotics.
4. Theorem 1 excludes a neighborhood of the cooperative equilibrium V_0 via the parameter r ; matching analysis between cooperative-regime convergence and congestion-regime repulsion is not provided.
5. Domain-specific application requires operationalising slack x , load V , capacity Q , the friction coupling, and ruling out the competing mechanisms of Section 7.

The rolling-window variance estimator used in empirical applications *under*-estimates the stationary variance near the fold (by up to $\approx 2\times$ at $\mu = 0.02$ in the simulation of Appendix B.5), precisely the pre-transition regime in which the variance early-warning signal is read; empirical users must either calibrate the window length against system relaxation times or replace the rolling-window estimator with an AR(1)-based variance reconstruction, which corrects the relaxation-time under-estimate.

A candidate embedding of this local model into a wider verification-geometric framework is developed in a separate document; the present paper does not depend on that embedding for any of its theorems or empirical predictions.

8.3 Distinguishing Predictions and Scope of Novelty

What this framework predicts that standard saddle-node CSD does not (with reference points [5, 6] in the standard CSD literature).

1. *Friction-divergence exponent $-1/2$ under inverse coupling.* Standard CSD predicts critical slowing down and variance inflation near a saddle-node but is silent on per-unit coordination cost. Proposition 2 derives $F \sim (Q - V)^{-1/2}$ as a conditional prediction tied to the structural assumption $F = k/x$. A different coupling $F = k/x^a$ would give exponent $-a/2$, so the specific value is empirically informative about coupling form.
2. *Friction-barrier inadmissibility.* Proposition 3 and Criterion 1 establish that, on the reduced slow flow, endogenous dynamics are *repelled* from the fold in the congestion regime. Standard CSD treats the fold as a destination of slow drift; this framework treats it as accessible only under exogenous forcing past a parameter-dependent barrier.
3. *Cross-observable consistency (σ^2/τ flatness).* The two CSD observables co-scale with exponent $-1/2$ in the present framework, so their ratio is μ -independent. A systematic μ -dependence of σ^2/τ falsifies the constant-intensity-noise assumption *before* it falsifies the fold classification.
4. *Three-model falsification structure.* The M1/M2/M3 design distinguishes between “no divergence,” “fold-exponent divergence,” and “generic power-law divergence.” Standard CSD diagnostics typically report only the presence of slowing down rather than the discrimination between exponents.

What this framework does not claim to do better than standard CSD. Detection rates for approaching transitions; sensitivity to noise-driven near-bifurcation behaviour; classification at the fold itself (open, see §6); behaviour beyond the inverse-coupling structural assumption $F = k/x$. (The $\delta_{\min}(r)$ trade-off curve is not a distinguishing prediction: the threshold is an artifact of the Lyapunov normalization rather than a property of the dynamics; see Remark 5.)

Pre-registered failure conditions. Apply the protocol of Appendix A to an observational dataset for which: $\mu = Q - V$ is operationalisable with proxies meeting the standards of §A.6; $n \geq 30$ paired (μ_i, τ_i) observations are available; $\text{CV}_\mu \leq 0.20$ and $\text{CV}_\tau \leq 0.20$; the system is in a pre-transition regime with monotone-decreasing μ .

Saddle-node classification falsified if any of the following hold:

- (i) M2 (linear, no divergence) is preferred over M1 by both AIC and BIC, with corroboration per A.3.

- (ii) A Wald test of H_0 : slope $= -1/2$ rejects at $\alpha = 0.05$ on the log-log fit of τ vs μ in two independent datasets meeting the protocol preconditions (or in two operationalisations of the same series; §A.6).

Conditions (i) and (ii) falsify the saddle-node classification of the local mechanism.

Test-voiding condition (variance branch). If the σ^2/τ log-log slope is significantly non-zero at $p < 0.01$ in both of two independent perturbation subsets of the same series, the constant-noise-intensity assumption underlying the OU *variance* has failed, which voids the variance branch of the test rather than falsifying the classification. Because $\tau = 1/\lambda$ carries no diffusion constant, the recovery-time branch remains admissible, and classification inference is then restricted to the τ branch.

The two-dataset requirement in (ii) applies symmetrically to support: a claim of fold-consistent support likewise requires the $-1/2$ exponent to replicate across two independent datasets (or two operationalisations of one series; §A.6), rather than a single near $-1/2$ fit, since operationalisation error in the μ proxy can manufacture spurious support as readily as spurious rejection. The symmetry is deliberate: a protocol resting on falsifiability discipline should not admit support on a weaker evidential gate than it demands for rejection. Single-dataset results are reported as suggestive rather than confirmatory, and the competing-mechanism checks of §A.3 apply on top of the two-dataset gate rather than in place of it. The classification falsifiers (i) and (ii) do not separately falsify the friction-coupling assumption $F = k/x$ (Proposition 2), which the CSD protocol does not directly test. A test of the coupling form requires direct measurement of per-unit coordination cost F across the observation window and a log-log fit of F vs μ ; the predicted slope is $-a/2$ under $F = k/x^a$, with $a = 1$ being the framework’s structural assumption. Falsification of $a = 1$ does not falsify the saddle-node classification; the two failure modes are tested by separate empirical operations, and either may obtain without the other.

Appendix A: Empirical Protocol for Testing P4

A.1 Purpose

A protocol for testing $\tau \propto |\mu|^{-1/2}$ and $\text{Var} \propto |\mu|^{-1/2}$ near a capacity-constrained transition, where $\mu = Q - V$.

A.2 Variable Mapping

Control parameter μ : the margin between capacity and load, $\mu = Q - V$. Recovery time τ : time to return to pre-perturbation state after a small shock. Variance σ^2 : variance of the system’s key output metric over a rolling window.

A.2b The Assumed Stochastic Model and the Observational Clock

The inference of §A.3 is exact under the following explicit model, stated here so the protocol is reproducible without reverse-engineering it from the estimands. Near the attracting branch $x^*(\mu) = \sqrt{\mu}$, the deviation $Y_t := x_t - x^*(\mu)$ is modelled by the linearized Ornstein–Uhlenbeck equation

$$dY_t = -\lambda(\mu) Y_t dt + \sqrt{D} dW_t, \quad \lambda(\mu) = 2\sqrt{\mu},$$

with λ the fold linearization rate of Proposition 1 and $D > 0$ a constant noise intensity. Its stationary statistics are the protocol’s estimands: recovery (autocorrelation) time $\tau(\mu) = 1/\lambda = 1/(2\sqrt{\mu})$, stationary variance $\sigma^2(\mu) = D/(2\lambda) = D/(4\sqrt{\mu})$, and the μ -independent cross-check $\sigma^2/\tau = D/2$ (§7). The *observational clock*: all times are in the fast variable’s units; the sampling interval Δ and estimation window T_w must satisfy $\Delta \ll \tau(\mu) \ll T_w$, and μ is treated as quasi-static over each window, i.e. the exogenous drive timescale exceeds T_w (windows violating this are excluded by §A.6’s identification checks). Under this model the log-log slopes of §A.3 are exactly $-1/2$ for both τ and σ^2 against μ ; departures from OU stationarity (time-varying D , heavy tails, oscillatory recovery) are competing-mechanism flags handled in §A.3, not fit parameters.

A.3 Primary Inference: Log-Log Regression

Under multiplicative observation noise, $y_{\text{obs}} = y_{\text{true}}(1 + \epsilon)$ with ϵ small, $\log y_{\text{obs}} \approx \log y_{\text{true}} + \epsilon$. The log-log regression of an observable against μ is therefore correctly specified: the slope estimates the exponent directly, with approximately Gaussian residuals.

For τ and σ^2 , the fold predictions in log-log space are

$$\log \tau = a_\tau - \frac{1}{2} \log \mu, \quad \log \sigma^2 = a_\sigma - \frac{1}{2} \log \mu.$$

Fit by OLS in log-log coordinates:

$$\hat{s} = -\widehat{\text{slope}}(\log y_{\text{obs}} \text{ vs } \log \mu_{\text{obs}}),$$

with the 95% CI obtained from the regression standard error on the slope.

Model comparison. Fit three models:

- M1 (fold): slope constrained to $-1/2$; intercept by OLS in log-log space.
- M2 (linear, no divergence): $y = c\mu + d$ in linear-linear space.
- M3 (free exponent): slope unconstrained in log-log space.

Inference structure. M1 vs. M3 is a nested comparison (M1 imposes $\text{slope} = -1/2$). The appropriate test is a Wald test on the log-log slope: reject H_0 : $\text{slope} = -1/2$ if $|\widehat{\text{slope}} + 1/2|/\text{SE}(\widehat{\text{slope}}) > z_{1-\alpha/2}$ at the chosen significance level α , equivalently if the $(1 - \alpha)$ CI for the slope excludes $-1/2$. M1 vs. M2 is non-nested (different functional forms); compare via AIC/BIC evaluated on the observable in its original space (residuals $y_{\text{obs}} - \hat{y}$, not $\log y_{\text{obs}} - \log \hat{y}$), so both models are scored on the same loss. For confirmatory M2-preference the AIC/BIC ranking is corroborated by out-of-sample predictive loss or a Vuong non-nested test; the original-space AIC/BIC is otherwise descriptive (B.2).

Decision rule:

- M2 dominates M1 by both AIC and BIC, corroborated per the inference-structure paragraph $\Rightarrow$ P4 falsified (no fold-consistent divergence).
- Wald test fails to reject H_0 at $\alpha = 0.05$ AND M1 preferred over or comparable to M2 by AIC/BIC $\Rightarrow$ fold-consistent scaling supported in this dataset (suggestive; confirmatory support requires two-dataset replication, §8.3).
- Wald test rejects H_0 at $\alpha = 0.05 \Rightarrow$ saddle-node classification falsified at this significance level; the framework's CSD prediction is rejected for this dataset. The result does not by itself identify the alternative bifurcation class (Hopf gives slope -1 , cusp gives a different exponent), nor does it speak to the friction-coupling form a in $F = k/x^a$, which the protocol does not directly test.

Cross-check. Under P4, σ^2/τ is constant in μ . Plot $\hat{\sigma}_i^2/\hat{\tau}_i$ against μ_i in log-log space and test for slope $= 0$. Systematic dependence on μ falsifies the constant-noise-intensity assumption underlying the OU derivation, even if the individual scaling fits look consistent.

Competing-mechanism checks. Before accepting M1: confirm the 95% CIs of $\hat{s}_\tau$ and $\hat{s}_\sigma$ both contain $1/2$ and are well separated from 1 (Hopf gives slope magnitude 1); verify monotone rather than transient-amplified recovery (nonnormality check); estimate dominant timescales and check they do not match system delays (delay check).

Alternative linearisation forms. The algebraically equivalent forms $\tau^{-2} = a\mu + b$ and $\sigma^{-4} = a'\mu + b'$ acquire β -dependent heteroscedasticity under multiplicative noise on the observable, leading to anti-conservative profile-likelihood CIs. The log-log form above is preferred for inference; the inverse-square forms remain useful for visual inspection because they map fold-consistent data to straight lines in linear-linear axes.

Caveat on the collapse forecast. The M1 fit yields a forecast collapse location as the margin at which the fitted curve reaches a chosen recovery-time threshold τ_c : $\hat{\mu}^*(\tau_c) = \exp(2(a_\tau - \log \tau_c))$ under the fold

slope $-1/2$. The threshold must be stated: the form $\exp(2a_\tau)$ corresponds to $\tau_c = 1$ in whatever units τ is measured, a hidden unit convention. This is a point estimate with intrinsic identifiability limits: the observed Q is a proxy for the true dynamical capacity, subject to systematic measurement bias, and a μ^* derived from the fit absorbs that bias along with the model. The decomposition $Q_{\text{true}} = Q_{\text{observed}} + \text{bias}$ is *internal to the protocol* and cannot, by itself, distinguish structural slack from regression artifact. Independent evidence on the proxy error (e.g., an audit measurement of true capacity, or a controlled perturbation experiment) is required to license a structural reading of $\hat{\mu}^*$. Without that evidence, treat $\hat{\mu}^*$ as a calibrated warning rather than a forecast.

A.4 Protocol Steps

1. Operationalise $\mu = Q - V$; verify monotonicity over the observation window.
2. Measure τ from perturbation events. Minimum $n \geq 10$ (CI coverage is approximately nominal from $n = 10$ at the noise levels of Appendix B; Table 5); $n \geq 30$ recommended for discriminating power against M2.
3. Plot $\log \tau_i$ vs. $\log \mu_i$. If σ^2 data are also available, plot $\log \sigma_i^2$ vs. $\log \mu_i$ in parallel.
4. Fit M1, M2, M3 per A.3.
5. Apply the three-way decision rule, the σ^2/τ flatness cross-check, and the competing-mechanism checks.
6. If M1 is supported in this dataset (suggestive pending two-dataset replication, §8.3), report $\hat{s}_\tau$ and $\hat{s}_\sigma$ with regression 95% CIs and the cross-check slope. Report $\hat{\mu}^*$ with the structural-reading caveat from A.3.

Independence of recovery-time observations. The Wald inference of A.3 assumes the residuals of the log-log fit are uncorrelated. For τ , this requires the perturbation events generating successive (μ_i, τ_i) pairs to be independent. When perturbations are separated by at least one relaxation time and the system returns to baseline between events, the assumption holds. When the system exhibits slow trends, autocorrelated state, or perturbations spaced more closely than the relaxation time, recovery-time observations may be serially correlated and the OLS standard errors are biased downward (anti-conservative on the Wald test). Two corrections: (a) cluster-robust standard errors with clustering by perturbation epoch; or (b) block bootstrap with block length matched to the estimated autocorrelation timescale. Report the larger of the OLS and corrected CIs.

A.5 Variance-Only Variant and Autocorrelation Warning

If recovery-time data are unavailable, the variance test alone can be used. Rolling-window variance estimates are serially correlated by construction: adjacent windows share $w - 1$ observations. OLS standard errors on $\sigma^{-4} = a'\mu + b'$ are therefore biased downward, and significance tests are anti-conservative. Two corrections: (a) Newey–West standard errors with bandwidth equal to w ; or (b) subsample at interval w so windows do not overlap, reducing effective sample size by a factor of w . If the scaling result survives both corrections, it is robust. Recommended window length $w \in [5, 10]$, sensitivity-checked against $w = 3$ and $w = 15$.

A.6 Identification and Operationalisation Standards

The regression estimates a statistical association between μ and the CSD observables rather than a causal structural law. The structural interpretation (fold scaling) requires that variation in μ is not driven by an unobserved common cause that independently affects τ or σ^2 . In domains where controlled perturbation experiments are feasible, exogenous variation in μ can be introduced directly. In observational settings, the assumption that μ is the dominant channel should be stated and its plausibility assessed domain by domain.

Because Q (capacity) and V (load) are rarely directly observable, the operationalisation of these proxies is itself a methodological commitment that must be discharged before the protocol is run. Without minimum standards the test risks measurement-construct circularity: the operationalisation choice may itself be informed by the prediction being tested, and the slope estimate $\hat{s}$ inherits that dependence.

Minimum operationalisation standards.

1. *Pre-registration.* The proxies for Q and V must be specified, with the source data identified, before the data are examined. Adjusting the proxies after seeing $\hat{s}$ collapses the falsification structure.
2. *Independence from the framework’s predictions.* The proxy for Q must not be defined by reference to past episodes of the system’s collapse (e.g., $Q \neq V_{\text{observed-at-collapse}}$). The proxy for V must not invoke quantities defined by the friction coupling under test. Both proxies must be expressible without invoking the model whose predictions the protocol is testing.
3. *Alternative-operationalisation robustness.* At least one alternative proxy choice must be specified in advance and the protocol run on both. If the fold mechanism holds, $\hat{s}$ from the two operationalisations should overlap in 95% CI; both contain $-1/2$ or both exclude it. Sensitivity of $\hat{s}$ to the proxy choice is evidence against the structural interpretation, independent of the Wald-test outcome.

Standard (2) catches direct dependence between the proxy specification and the framework’s predictions. In domains where the framework’s vocabulary has already entered conventional measurement practice, indirect dependence may persist even when no explicit reference to the framework appears in the proxy definition; in such cases, operationalisations should be defensible without invoking the framework, and standard (3) becomes the primary safeguard against indirect circularity.

The collapse forecast $\hat{\mu}^*$ is a point estimate with uncertainty. It should not be treated as a precise prediction of collapse timing but as a structural warning: the system’s dynamics are consistent with a fold at this distance, and the margin is shrinking.

Appendix B: Synthetic Validation

B.1 Setup

Synthetic observations are generated from the OU linearisation of the fast subsystem with $Q = 1$, $D = 1$. For $i = 1, \dots, n = 40$, μ_i is swept linearly from 0.70 down to 0.02; the true CSD observables are $\tau_{\text{true}}(\mu) = 1/(2\sqrt{\mu})$ and $\text{Var}_{\text{true}}(\mu) = D/(4\sqrt{\mu})$. Multiplicative observation noise is added with $\text{CV}_{\mu} = 0.08$, $\text{CV}_{\tau} = 0.12$, $\text{CV}_{\sigma} = 0.15$.

B.2 Log-Log Inference (Primary)

For the pinned seed of Figure 2 (single realisation, $n = 40$; seed 10426 of the committed pipeline `fold_paper/table3_pipeline.py`):

Observable	$\hat{s}$	95% CI	ΔBIC_{M2-M1}	ΔBIC_{M3-M1}	R^2 (M1)
τ	0.47	[0.42, 0.52]	+104.1	+11.0	0.960
σ^2	0.47	[0.42, 0.53]	+50.2	−3.0	0.880

Table 3. Single-realisation log-log inference. Both 95% CIs contain the fold prediction $s = 1/2$. All six numeric cells are regenerated from the single pinned seed of Figure 2 (seed 10426; script committed beside the figure code); the τ row matches panel (a) to displayed precision. M2 (linear, no divergence) was rejected for both observables in that run. The ΔBIC_{M3-M1} column is realization-noisy by construction: across the 200 seeds of B.3 its distribution for τ is centred near zero (mean $\approx +1$, $sd \approx 7$, with $\approx 10\%$ of seeds below -8), because original-space BIC scoring after log-space fitting is a noisy instrument for the nested M1-vs-M3 comparison. The displayed seed’s +11.0 for τ is an upper-range draw from that ensemble, not evidence for M1 on its own; the Wald test on the log-log slope (A.3) is the primary inference for that comparison. The same pseudo-likelihood caveat applies to the M1-vs-M2 comparison: BIC differences after log-space fitting are not a proper likelihood ratio, so M2 rejection is corroborated here by the instruments A.3 mandates, computed on the same seed. Out-of-sample predictive loss (5-fold on the (μ_i, y_i) pairs, original-space MSE): τ , M1 0.017 vs M2 0.474; σ^2 , M1 0.010 vs M2 0.071. Vuong non-nested test (M1 vs M2): τ , $z = 4.27$, $p < 10^{-4}$; σ^2 , $z = 2.69$, $p = 0.007$. Reading: both instruments corroborate the BIC ranking – M1 dominates M2 out of sample and by pointwise log-likelihood, for both observables.

Cross-check. The ratio σ^2/τ regressed on $\log \mu$ has log-log slope -0.005 with $p = 0.88$, consistent with the fold prediction that the ratio is μ -independent.

B.3 Robustness Across Seeds

To convert the single-seed result into a statement about protocol operating characteristics, the validation was repeated for 200 independent seeds at $n = 40$:

Observable	$\widehat{s}$	$\text{sd}(\widehat{s})$	CI coverage of 1/2
τ	0.496	0.022	98.0%
σ^2	0.499	0.028	96.5%

Table 4. Distribution of log-log slope estimates across 200 seeds. CI coverage is consistent with the nominal 95% rate within Monte Carlo error (τ : 98.0%, Wilson 95% interval [95.0%, 99.2%]; σ^2 : 96.5%, [93.0%, 98.3%]); the independent 300-replication sweep of B.4 sits at 94–95% across all n , likewise nominal.

B.4 Power

Coverage of the fold slope $-1/2$ inside the 95% CI as a function of n (300 replications per n):

n	$\Pr(\text{CI}_{95\%} \ni -1/2)$
10	94%
20	94%
30	95%
40	95%
60	94%
80	94%

Table 5. Log-log CI coverage as a function of sample size for the variance observable. Coverage is approximately nominal (94–95%) from $n = 10$ onward at these noise levels; the discriminating power against M2 (linear) saturates near $n \approx 30$.

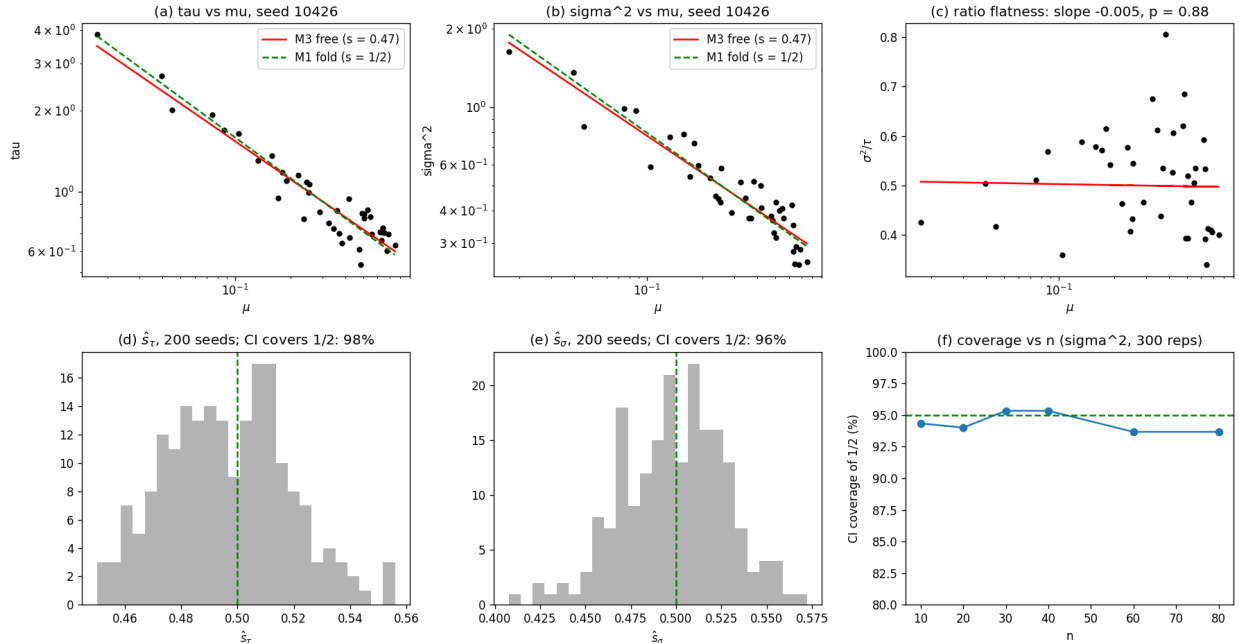


Figure 2. Log-log validation results. (a) τ vs. μ on log-log axes with the M3 free-slope fit (red) and the fold-constrained M1 fit (green dashed); both overlap visibly. (b) Same for σ^2 . (c) The σ^2/τ flatness cross-check on log-log axes; the log-log slope is consistent with zero. (d, e) Distribution of $\hat{s}_\tau$ and $\hat{s}_\sigma$ across 200 seeds; both centre on the fold prediction (green dashed), covering 1/2 at 98.0% and 96.5% respectively (Table 4; rounded to 98% and 96% in-panel). (f) CI coverage of the fold slope as a function of n , approximately nominal (94–95%) from $n \geq 10$.

B.5 Rolling-Window Estimator Bias (Caveat)

The protocol as written requires a stationary-variance estimate per μ_i ; in real data this is approximated by a rolling-window variance estimator. A direct OU simulation of the linearised fast subsystem ($T = 400$ time units per μ , sampling step 1 time unit, window $w = 8$, 200 seeds), compared against the analytic stationary variance $\text{Var}_{\text{true}} = D/(4\sqrt{\mu})$ of §7, shows that the rolling-window estimator carries a finite-sample bias that *under-estimates* the stationary variance, monotonically in μ : the ratio (estimator / Var_{true}) rises from ≈ 0.52 near the fold ($\mu = 0.02$, relaxation time $1/\lambda \approx 3.5$) to ≈ 0.94 far from it ($\mu = 0.70$, $1/\lambda \approx 0.6$), never exceeding 1. The under-estimate is largest near the fold because the relaxation time there exceeds the window span, so an 8-sample window cannot accumulate the full stationary variance; far from the fold the samples decorrelate and the estimate approaches the unbiased value. The effect is exact in closed form: for a discrete OU process with one-step autocorrelation $a = e^{-\lambda}$, the Bessel-corrected windowed sample variance satisfies $\mathbb{E}[\hat{V}]/\text{Var}_{\text{true}} = 1 - \frac{1}{w(w-1)} \sum_{i \neq j} a^{|i-j|} < 1$, increasing monotonically toward 1 as λ grows (Figure 3). The identity is the standard serial-correlation bias of the sample variance: writing $s^2 = \frac{1}{w-1} \sum_i (x_i - \bar{x})^2$ for a zero-mean stationary process with variance σ^2 and autocorrelation $\rho(k) = a^{|k|}$, one has $\mathbb{E}[\sum_i (x_i - \bar{x})^2] = (w-1)\sigma^2 - \frac{\sigma^2}{w} \sum_{i \neq j} \rho(|i-j|)$, whence $\mathbb{E}[s^2] = \sigma^2(1 - \frac{1}{w(w-1)} \sum_{i \neq j} a^{|i-j|})$. Evaluated at $w = 8$ and $\lambda(\mu) = 2\sqrt{\mu}$, the closed form gives:

μ	0.02	0.10	0.30	0.50	0.70
relaxation $1/\lambda$	3.54	1.58	0.91	0.71	0.60
ratio $\mathbb{E}[\hat{V}]/\text{Var}_{\text{true}}$	0.52	0.76	0.88	0.92	0.94

The simulation points of Figure 3 match these values to within seed noise. The validation of B.2-B.4 used the analytic stationary variance to isolate the inference method from this estimator bias; empirical applications combining the two should calibrate the window length w against the longest plausible relaxation time and report sensitivity across $w \in \{3, 5, 8, 15\}$, or replace the rolling-window estimator with an AR(1) coefficient ρ and back out $\text{Var} = \text{Var}(\varepsilon)/(1 - \rho^2)$, which removes the relaxation-time under-estimate near the fold.

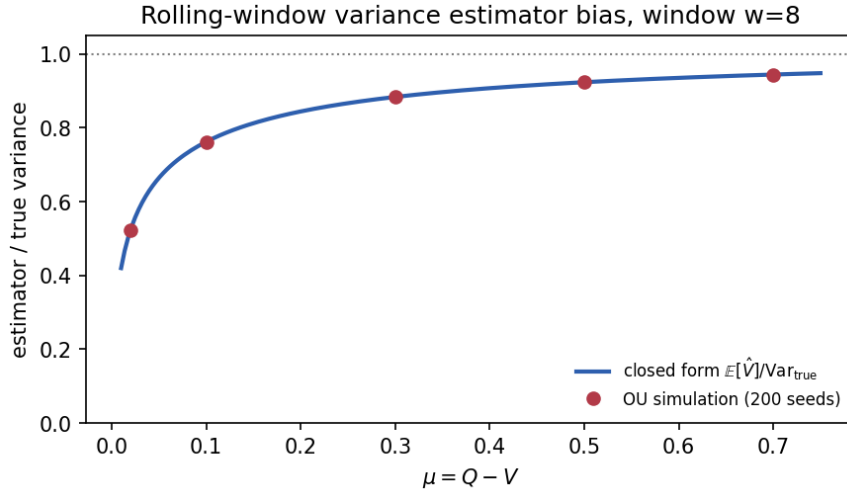


Figure 3. Finite-sample bias of the rolling-window variance estimator (window $w = 8$, sample step one time unit) relative to the OU stationary variance $D/(4\sqrt{\mu})$: the estimator under-estimates monotonically, most severely near the fold. Solid curve, closed-form $\mathbb{E}[\hat{V}]/\text{Var}_{\text{true}}$; points, OU simulation (200 seeds).

Appendix C: Full-System Simulations

C.1 Purpose and Pipeline

This appendix reports direct numerical integration of the full singular system (1)–(2) against every checkable clause of Theorem 1, Proposition 5, and Corollary 1. The simulations are *consistent with* the certificate; they do not extend it, and the certificate’s authority stays with the proof. All numbers below compile from a single deterministic execution of `appendix_c_pipeline.py` (tolerances pinned at $\text{rtol} = 10^{-10}$, $\text{atol} = 10^{-12}$), which writes `appendix_c_numbers.json`; nothing is hand-transcribed. *Reproducibility*: the pipeline (237 lines, SHA-256 prefix `b2c93cd5879788f5`) and its output JSON (SHA-256 prefix `7ca0c84431ca0bb6`) are archived alongside the source; one execution regenerates the JSON and Figure C1.

Parameters: $(\alpha, \beta, k, Q) = (1, 1, 1, 2)$, so $x_0 = 1$, $V_0 = 1$; tube parameters $\delta = 0.3$, $r = 0.2$. Derived constants: $\Gamma_\sharp = 7/3$, $\Gamma_b = 17/3$, $\gamma_0 \approx 0.1180$, $w \approx 112.02$, $L_- \approx 125.24$, certified rate $\kappa = \mu_0 \Gamma_\sharp \Gamma_b / 16 \approx 1.16$.

Two values of the tube half-width appear, and they play different roles. The certified check of C.2 verifies the theorem at $C = 1$, the value at which the pipeline assembles the six-term ε_0 (binding entry: $\varepsilon_{\text{Lip}} = 1.76 \times 10^{-6}$; the residual entry alone would allow 3.24×10^{-3}). The demonstration runs of C.3 use a wider $C = 2$ tube, which makes the fiber collapse and the face structure visible at plotting scale. Theorem 1 holds for each fixed C with its own $\varepsilon_0(C)$.

C.2 The Certified Run

The run that directly instantiates the theorem comes first. At $\varepsilon = 1.50 \times 10^{-6} \leq \varepsilon_0 = 1.76 \times 10^{-6}$ (the $C = 1$ assembled threshold), the tube-initialized trajectory satisfies every clause of the $C = 1$ certificate: it exits through the cooperative face only; it stays within the $C = 1$ fast fiber, $\max_t |y| = 6.1 \times 10^{-4} \leq \sqrt{\varepsilon} = 1.22 \times 10^{-3}$; $x \geq \delta/2$ throughout ($\min_t x = 0.849$); and $dW/dt < 0$ pointwise, with empirical decay rate $\kappa_{\text{emp}} := \sup_t (dW/dt)/(-\varepsilon) \approx 9.30$ against the certified $\kappa \approx 1.16$ — the Lyapunov bound is conservative by a factor of about eight inside its own regime.

C.3 Mechanism Demonstrations at $\varepsilon = 0.002$

At the reference $\varepsilon = 0.002$ — below the residual constraint $\varepsilon_{\text{res}} = 0.00324$ but well above the assembled ε_0 , so these are demonstrations of the mechanism and of the bound’s conservatism, *not* certified instances — all five tube-initialized trajectories in the $C = 2$ tube satisfy every clause: exit through the cooperative face only; $|y| \leq C\sqrt{\varepsilon}$ throughout; $x \geq \delta/2$ throughout; $dW/dt < 0$ pointwise, with $\kappa_{\text{emp}} \approx 9.29$ – 9.45 . The pipeline’s aggregate flag `theorem1_all_pass` is true.

Off-tube initial conditions ($|y_0| > C\sqrt{\varepsilon}$) collapse onto the tube on the fast time scale and then obey the same clauses. These runs are illustrative only: the theorem does not claim them.

C.4 Exit-Time Law

Theorem 1 bounds the exit time by $(M - L_- + c_T \varepsilon)/(\kappa \varepsilon)$, an $O(1/\varepsilon)$ law. Across $\varepsilon \in [5 \times 10^{-4}, 3 \times 10^{-3}]$ the measured exit times give $T \cdot \varepsilon \in [1.868, 1.874]$: the $1/\varepsilon$ scaling holds to within 0.4% over the sampled range, with the certified bound conservative by a constant factor. The certified run of C.2 obeys the same law at its own scale ($T_{\text{exit}} \approx 3.67 \times 10^5$).

C.5 Figure

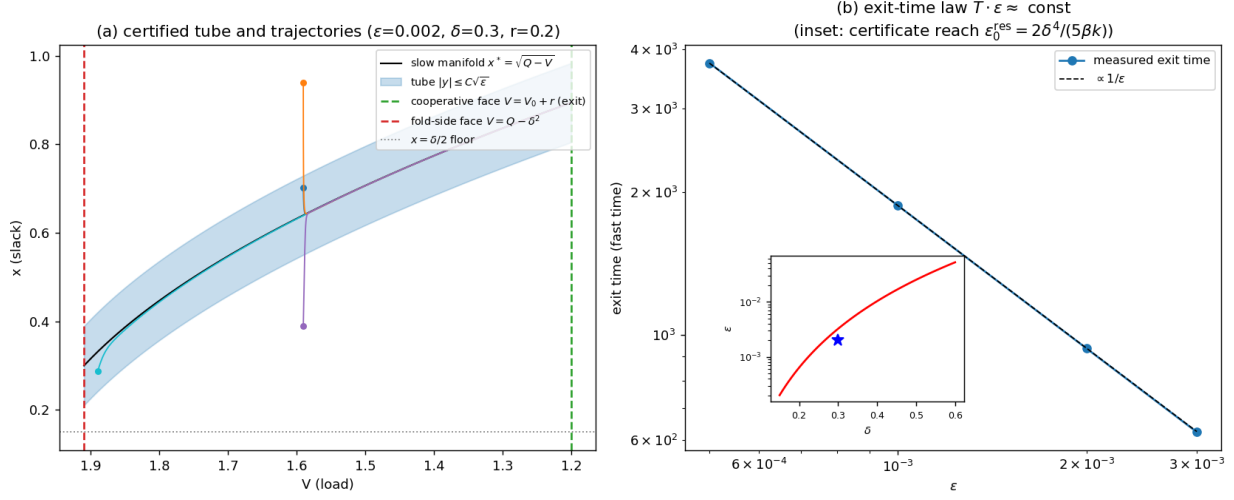


Figure C1. Left: the slow-manifold tube $T_{C,\varepsilon}(r, \delta)$ with its faces, tube-initialized trajectories (certificate clauses checked pointwise), and off-tube trajectories collapsing onto the tube. Right: exit-time scaling across ε (log-log), with the reach-boundary inset. Generated by `appendix_c_pipeline.py`.

Appendix D: Explicit Thresholds for Theorem 1

D.1 The Six Entries

With the structural constants of Table 2 and $u_+ = \sqrt{x_0^2 - r}$:

$$\begin{aligned} \varepsilon_{\text{tube}} &:= \frac{\delta^2}{4C^2}, & \varepsilon_{\text{res}} &:= \frac{2\delta^4}{5\beta k}, & \varepsilon_{\text{Lip}} &:= \left(\frac{\gamma_0(r)}{2L_g C} \right)^2, & \varepsilon_1 &:= \left(\frac{\Gamma_{\#}(\delta) \Gamma_b(\delta) \delta^3}{12 C (\beta k)^2} \right)^2, \\ \varepsilon_x &:= \left(\frac{r}{C(x_0 + u_+)} \right)^2, & \varepsilon_{\text{win}} &:= \min \left\{ \frac{16 w k \delta^5}{3(\beta k)^2}, \left(\frac{4 w k \delta^2}{3 \beta k C} \right)^2, \frac{C^2 u_+^4}{\gamma_0(r)^2} \right\}, \\ \varepsilon_0 &:= \min\{\varepsilon_{\text{tube}}, \varepsilon_{\text{res}}, \varepsilon_{\text{Lip}}, \varepsilon_1, \varepsilon_x, \varepsilon_{\text{win}}\}. \end{aligned}$$

Each entry enforces one bound in the proof of Theorem 1; the Assembly step of §5.4 states the correspondence, one bound per entry. With the tube half-width C substituted, these are the values assembled by `appendix_c_pipeline.py` (Appendix C).

D.2 Window Nonemptiness

When the window cap of Theorem 1 is $L_- + C^2/2 - c_M \sqrt{\varepsilon}$, no extra condition is needed: the window is nonempty because $2c_M \sqrt{\varepsilon} \leq C^2/2$ is exactly the third entry of ε_{win} . When the cap is $wk/\delta - c_\delta \varepsilon$ (the fold-cap case), Theorem 1 assumes the explicit fold-gap condition $c_M \sqrt{\varepsilon} + c_\delta \varepsilon < wk(1/\delta - 1/u_+)$; a sufficient closed form is

$$\varepsilon \leq \min \left\{ \left(\frac{wk(1/\delta - 1/u_+)}{2c_M} \right)^2, \frac{wk(1/\delta - 1/u_+)}{2c_\delta} \right\},$$

each factor forcing its term below half the gap. This makes Lemma 1’s “gap minus $O(\varepsilon)$ ” explicit (see the Remark after Lemma 1): for δ near u_+ the fold gap is small and this is a genuine additional smallness requirement that the six-term ε_0 does not enforce.

D.3 Reduction at $C = 1$

At $C = 1$ the C -carrying entries reduce to $\varepsilon_{\text{tube}} = \delta^2/4$, $\varepsilon_{\text{Lip}} = (\gamma_0/(2L_g))^2$, and $\varepsilon_x = (r/(x_0 + u_+))^2$, the forms used in the assembled ε_0 of Appendix C (machine-verified term-by-term; `v70_lemmachain/verify_L7_final.py`).

Appendix E: Kernel-Verified Certificate Layer (Lean 4)

Seventeen finite obligations were staged from this paper into a verifier-bounded proof-search pipeline (Math-Solver; architecture and provenance discipline documented separately): ten from the certificate and estimand algebra (§3, §5, Prop. 6, A.2b/A.3, B) and seven sequenced from Theorem 1’s certificate chain (§5, §6.2, Appendix D), each instantiated at this paper’s constants. Every staged statement carries a declared claim card (completion criteria, edge-case checklist – Lean semantics such as $t/0$, `zpow` at 0, natural subtraction are declared, not discovered – and excluded partial results), was ground-truthed numerically or by enumeration before staging, and passed a two-model adversarial statement-adequacy audit with recorded human closures (all seventeen terminally adjudicated).

Status. Sixteen obligations are proved in Lean 4 and kernel-checked, as closed terms with no placeholders:

Lean name	This paper	Content
<code>chain_completed_square</code>	Lemma 3 (§6.2)	the completed-square display W
<code>chain_g_at_up</code>	§5 / App. D	exact $g(u_+)$ evaluation
<code>chain_b_at_up</code>	§5 / App. D	$ b(u_+) $ identity
<code>chain_fold_floor</code>	Thm. 1 certificate	floor inequality at $u = \delta$, explicit constants
<code>chain_coop_floor</code>	Thm. 1 certificate	floor inequality at $u = u_+$, $c_T = g_0^2/(32u_+^4)$
<code>fold_eigenvalue</code>	Prop. 1 / Prop. 6 rate	exact derivative $\partial_x(\mu - x^2)$ at the branch
<code>fold_standoff</code>	§5	standoff identity $a - bk/(bk/a) = 0$, all real a
<code>fold_loglog_slope</code>	A.2b/A.3	$\log(1/(2\sqrt{\mu})) = -\log 2 - \frac{1}{2}\log \mu$ (the $-1/2$ slope)
<code>fold_ratio_constant</code>	A.2b	$\sigma^2/\tau = D/2$ cross-check
<code>chain_exit_time</code>	Thm. 1 / App. D (L7)	exit bound $T \leq (M - L_- + c_T\varepsilon)/(\kappa\varepsilon)$
<code>fold_fast_equilibria</code>	§6 / Prop. 1	equilibria of the desingularized fast field $x(\mu - x^2)$
<code>fold_friction_scaling</code>	Prop. 2	$F(x^*) = k\mu^{-1/2}$ on the attracting branch
<code>fold_bernoulli_identity</code>	§6	Bernoulli linearization $z' = 2 - 2mz$ for $z = x^{-2}$
<code>fold_weight_balance</code>	§6 chart	$(1, 2, 4)$ quasi-homogeneous balance: $x\mu$ and x^3 scale as $\varepsilon^{3/4}$
<code>fold_homog_decay</code>	§6 (σ_0 clause)	homogeneous exponent decay beyond $\sigma_0 = -2\bar{\mu}_0/(\beta k)$
<code>fold_rw_bias_pairs</code>	App. B.5	rolling-window bias pair sum at $w = 8$

The $u = u_+$ floor’s provenance is worth a sentence: the first kernel-accepted proof of it (a model-produced route) was withheld by the pipeline’s anomaly tripwire as a statistical outlier; a subsequent refinement pass proved the SAME statement by naming the completed-square gap directly, at a complexity statistic seven times lower, and that proof is the retained carrier. The tripwire flagged an expensive proof, not a wrong statement – which is what such a gate is for. The remaining one, the window-cap biconditional, is open as a formalization target, independent of its prose proof here: it has now rejected across six consecutive search-tier runs (two model families), and the pipeline’s next step for it is a statement-adequacy audit of the staged formalization rather than further search.

What this layer does and does not claim. Each certificate proves exactly the finite algebraic proposition on its claim card, instantiated at this paper’s constants: a third verification tier alongside the analytic proofs and the computer-algebra audit scripts. It does not touch the invariant-manifold arguments of §6.2, the open gluing, or any asymptotic statement. The nine certificates of the 2026-07-13 set additionally survived an adversarial pass in which the pipeline attempted to verify each statement’s negation and to close each statement by `trivial` alone: no negation verified, and none of the nine is trivially closable. The seven later certificates await the same pass. One dependency is itself machine-measured: the formal $u = \delta$ floor proof

reuses the separately proved completed-square lemma, the reuse read off the kernel proof term’s dependency surface – mirroring, in the formal layer, exactly the role that identity plays in Lemma 3.

Basis. All sixteen terms are closed (no metavariables, no `sorry`); the kernel-transitive closure of every term carries at most the classical trio (`propext`, `Classical.choice`, `Quot.sound`); on the direct dependency surfaces no axiom constants appear except `propext` on the exit-time proof. Toolchain: Lean 4.30 + pinned `mathlib`, unchanged across the runs. The seven proofs added after the 2026-07-13 set were produced on 2026-07-20 (pipeline runs 14, 16, and 18): six by a single frontier-model arm and one, the exit-time bound, in the same campaign, each retained only after statement conformance against the declared type, a clean axiom inventory, and structural checks on the proof term. The anomaly tripwire fired once more in this era: a model re-proof of the $u = u_+$ floor was withheld at a step-label entropy of 3.68 against the calibrated cap of 3.2 while the original retained carrier stood. Statements, claim cards, proofs, run records, and the retained-proof registry are in the MathSolver corpus beside this paper (`mathsolver_mvp/`; runs `run_1783483932`, `run_1783629534`, `run_1783638352`, `run_1783715068`).

References

- [1] S. Strogatz, *Nonlinear Dynamics and Chaos*, Westview Press, 2015.
- [2] J. Guckenheimer and P. Holmes, *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields*, Springer, 1983.
- [3] N. Fenichel, “Geometric singular perturbation theory for ordinary differential equations,” *J. Differential Equations* 31 (1979), 53–98.
- [4] M. Krupa and P. Szmolyan, “Extending geometric singular perturbation theory to non-hyperbolic points: fold and canard points in two dimensions,” *SIAM J. Math. Anal.* 33 (2001), 286–314.
- [5] M. Scheffer, J. Bascompte, W. A. Brock, et al., “Early-warning signals for critical transitions,” *Nature* 461 (2009), 53–59.
- [6] C. Boettiger and A. Hastings, “Quantifying limits to detection of early warning for critical transitions,” *J. R. Soc. Interface* 9 (2012), 2527–2539.
- [7] E. H. van Nes and M. Scheffer, “Slow recovery from perturbations as a generic indicator of a nearby catastrophic shift,” *American Naturalist* 169 (2007), 738–747.
- [8] H. Held and T. Kleinen, “Detection of climate system bifurcations by degenerate fingerprinting,” *Geophysical Research Letters* 31 (2004), L23207.
- [9] S. R. Carpenter and W. A. Brock, “Rising variance: a leading indicator of ecological transition,” *Ecology Letters* 9 (2006), 311–318.
- [10] C. K. R. T. Jones, “Geometric singular perturbation theory,” in *Dynamical Systems (Montecatini Terme, 1994)*, Lecture Notes in Math. 1609, Springer, 1995, 44–118.
- [11] C. Kuehn, “A mathematical framework for critical transitions: bifurcations, fast–slow systems and stochastic dynamics,” *Physica D* 240 (2011), 1020–1035.
- [12] C. Kuehn, *Multiple Time Scale Dynamics*, Applied Mathematical Sciences 191, Springer, 2015.
- [13] C. Kuehn, “Normal hyperbolicity and unbounded critical manifolds,” *Nonlinearity* 27 (2014), 1351–1366.
- [14] N. Berglund and B. Gentz, *Noise-Induced Phenomena in Slow–Fast Dynamical Systems*, Springer, 2006.
- [15] M. Wechselberger, *Geometric Singular Perturbation Theory Beyond the Standard Form*, Frontiers in Applied Dynamical Systems 6, Springer, 2020.
- [16] P. Ashwin, S. Wiczeorek, R. Vitolo, and P. Cox, “Tipping points in open systems: bifurcation, noise-induced and rate-dependent examples in the climate system,” *Phil. Trans. R. Soc. A* 370 (2012), 1166–1184.

- [17] S. Prajna and A. Jadbabaie, “Safety verification of hybrid systems using barrier certificates,” in *Hybrid Systems: Computation and Control*, Lecture Notes in Comput. Sci. 2993, Springer, 2004, 477–492.
- [18] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, “Control barrier functions: theory and applications,” in *Proc. 18th European Control Conference (ECC)*, 2019, 3420–3431.
- [19] C. Wissel, “A universal law of the characteristic return time near thresholds,” *Oecologia* 65 (1984), 101–107.
- [20] L. Kleinrock, *Queueing Systems, Volume 1: Theory*, Wiley, 1975.
- [21] J. Sotomayor and M. Zhitomirskii, “Impasse singularities of differential systems of the form $A(x)x' = F(x)$,” *J. Differential Equations* 169 (2001), 567–587.
- [22] M. Zhitomirskii, “Local normal forms for constrained systems on 2-manifolds,” *Bol. Soc. Bras. Mat.* 24 (1993), 211–232.
- [23] P. T. Cardin, P. R. da Silva, and M. A. Teixeira, “Implicit differential equations with impasse singularities and singular perturbation problems,” *Israel J. Math.* 189 (2012), 307–322.
- [24] P. T. Cardin and M. A. Teixeira, “A geometric singular perturbation theory approach to constrained differential equations,” *Math. Nachr.* 292 (2019), 892–904.
- [25] J. Borsotti, H. Jardón-Kojakhmetov, and M. Sensi, “Slow–fast dynamics in a planar parasite–host model with an extinction singularity,” arXiv:2602.19257 (2026).
- [26] K. U. Kristiansen and S. J. Hogan, “Regularizations of two-fold bifurcations in planar piecewise smooth systems using blowup,” *SIAM J. Appl. Dyn. Syst.* 14 (2015), 1731–1786.